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Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

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Electrical Engineering

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High-Density CMOS-based neural recording using Ultra-Flexible Tentacle Electrodes

Master's Thesis

Houman Javaheri
hjavaheri@ethz.ch

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Professor: Dr. Mehmet Fatih Yanik

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Abstract

The development of implantable neural probes has reshaped neuroscience, enabling large-scale, brain-wide recordings of neural population activity and leading to significant discoveries in brain research. However, existing electrophysiological recording systems are limited in their ability to achieve high channel counts without significantly increasing the size of implantable devices. In this work, we present a novel high-density neural recording system that integrates Ultra-Flexible Tentacle Electrodes (UFTEs) with CMOS-based microelectrode arrays (MEA), capable of capturing up to 1,024 channels simultaneously in rodent models. By combining the long-term stability and biocompatibility of UFTEs with the high-resolution recording capabilities of CMOS-MEAs, our system facilitates intracortical signal acquisition from multiple brain regions with a channel density of 158 channels/mm^2, enabling multi-thousand-channel recordings within a millimeter-scale area. Furthermore, we demonstrate the system's ability to record stable, single-unit activity in vivo. Additionally, we introduce a wireless version of the system, capable of recording in freely moving animals. This system represents an advancement in neural interfacing technologies, providing a scalable solution for high-density, long-term neural recordings in experimental neuroscience.

Declaration of Originality

I hereby confirm that I am the sole author of the written work here enclosed and that I have compiled it in my own words. Parts excepted are corrections of form and content by the supervisor. For a detailed version of the declaration of originality, please refer to Appendix A

Houman Javaheri,
Zurich, October 2024

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Chapter 1

Introduction

Vision, perception, and precise movement through environments are tasks that require the coordinated activity of multiple populations of neurons across different brain areas to function effectively [4][5][6]. Over the past decades, advancements in electrophysiology techniques and neuroelectronic implants have enabled a shift from single-cell recordings [7] to large-scale, brain-wide recordings [8], generating vast amounts of data that deepen our understanding of brain function. Moreover, the evolution of electrophysiological recording tools has led to the development of long-term, stable Brain-Computer Interfaces (BCIs) and neurotechnologies. These innovations offer promising therapeutic solutions for individuals with neurodegenerative conditions such as Amyotrophic Lateral Sclerosis (ALS) and Parkinson's disease by enabling precise neural control and interaction with external devices.

Our lab previously introduced Ultra-Flexible Tentacle Electrodes (UFTEs) [1], which can track the same population of neurons across sessions for several months with a 1.5-3 times higher signal-to-noise ratio compared to the state-of-the-art. Our ultra-flexible electrodes have multiple advantages for single-cell recording due to their minimal footprint, and no limitation in recording depth.

Integrated with eight Intan electrophysiology integrated circuits, these electrodes enabled us to record up to 512 channels with our neural logger. However, scaling up the number of simultaneously recorded channels by simply stacking multiple circuit boards becomes inefficient due to bulkiness, added weight, and limited scalability for implantation in freely moving animals.

On the other hand, the development of Complementary Metal-Oxide-Semiconductor (CMOS) microelectrode arrays (MEAs) has enabled the simultaneous recording of thousands of electrodes with sub-20 um pitch sizes [9][10], although their stiff, rigid form makes them unsuitable for brain implantation. In this project, we integrated our UFTEs with CMOS-MEAs [10] to combine the long-term flexibility of our electrodes with the

high-density recording capabilities of microelectrode arrays. We utilized a similar microstructure design to the previously introduced Flex2Chip interconnects [2], fabricating the interconnect between our flexible electrodes and the MEA1k.

The electrode design consisted of four bundles, with a customized number of contact pads per bundle based on the implantation location in the brain. The MEA-UFTE assembly was further secured using mechanical pressure to ensure stability over several weeks of recording. The results demonstrated a stable connection throughout the entire duration of in vivo recording. This system is capable of recording intracortical signals from 1,024 channels within a 2.1 mm by 3.85 mm area, significantly increasing the recording density compared to our previous Intan-based systems.

1.1. MEA1k: High-Resolution CMOS MEA Platform

The high-resolution CMOS microelectrode array (MEA) developed by Müller et al. [10] offers a powerful tool for studying neuronal activity at various levelsfrom subcellular to network scales. The device features 26,400 microelectrodes arranged with a 17.5 um pitch, spanning a large area of 3.85 x 2.10 mm^2 1.1. The array is reconfigurable, meaning that any 1,024 of the electrodes can be connected to low-noise readout channels, offering flexibility in selecting which neurons or neuronal processes to record.

The noise level of the recording system is particularly low, with input-referred noise in the action potential frequency band (300 Hz10 kHz) reported as 2.4 uV_{rms}, which is exceptionally low for this type of system. Such low noise is critical for detecting fine details in neuronal signals, such as action potentials or synaptic currents, which are often on the order of microvolts in amplitude. This allows the system to capture precise electrophysiological signals, even from small or distant neurons, with minimal interference.

Additionally, the device features three stages of amplification, providing a total programmable gain of up to 78 dB. This range of gain accommodates the wide variability in signal amplitude observed across different types of neural activity, making the MEA1k suitable for capturing both high-amplitude action potentials and low-amplitude local field potentials. Signals are digitized on-chip using 10-bit analog-to-digital converters (ADCs) at a sampling rate of 20 kHz, ensuring high temporal resolution. By digitizing the signals on-chip, interference from external noise sources is minimized, preserving signal integrity during off-chip transmission.

Furthermore, the device includes 32 programmable stimulation buffers that can deliver arbitrary voltage or current-controlled waveforms, offering flexible options for electrical stimulation experiments. The ability to simultaneously stimulate and record from the same array greatly enhances the system's versatility, making it applicable to a wide range of experimental paradigms, from network-level investigations to detailed studies of single-cell dynamics.

These CMOS MEAs have been successfully employed in several studies for in vitro neural recordings, offering valuable insights into bottom-up neuroscience approaches, such as in the investigation of autism-related neural development and the study of retinal organoid functionality [11][12]. However, the MEA1k is not suitable for in vivo applications due to its rigid structure. Nonetheless, recent advancements have demonstrated various methods for integrating flexible electrodes with the MEA1k chip, paving the way for more adaptable and scalable neural recording systems [2].

In our project, we integrated the CMOS-MEA platform with our Ultra-Flexible Tentacle Electrodes (UFTEs) to leverage the high-density recording capability of the MEA1k alongside the flexibility and long-term stability of UFTEs. This hybrid system allows for intracortical recordings from 1,024 channels within a compact 2.1 mm by 3.85 mm area, a significant improvement over previous Intan-based systems in both density and durability. This combination is particularly useful for chronic in vivo recordings, where both spatial resolution and long-term stability are paramount.

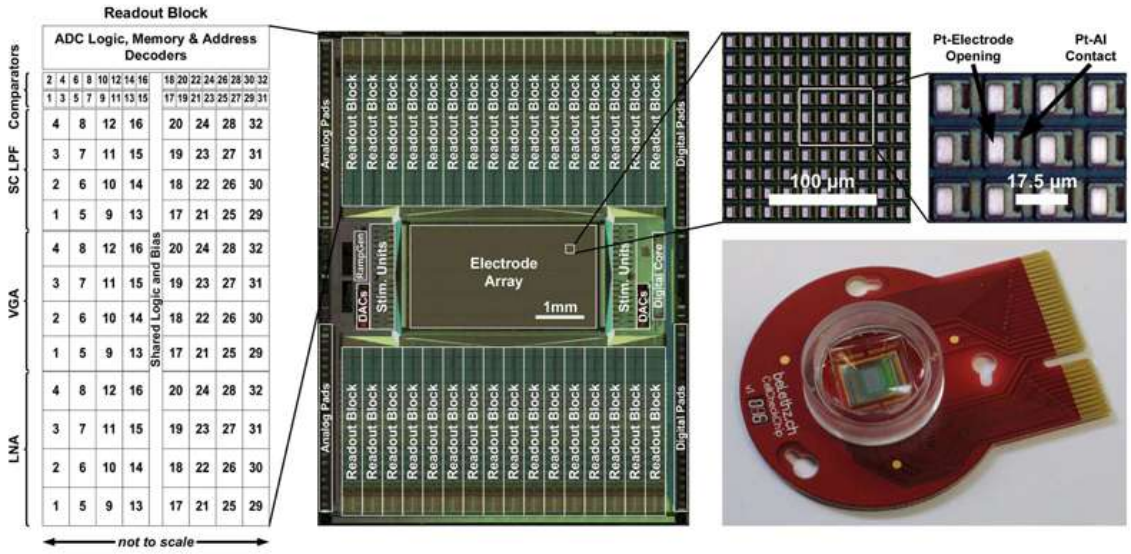


Figure 1.1.: MEA1k Overview, Layout of a readout block (32 channels), chip micrograph, close-up view of the electrode array and packaged device. The die size is 7.6 10.1 mm.

1.2. Ultra-Flexible Tentacle Electrodes (UFTEs)

Our lab recently developed and published the use of Ultra-Flexible Tentacle Electrodes (UFTEs) for long-term, stable tracking of neuronal ensembles spanning multiple brain areas [1]. These electrodes represent a significant advancement over existing technologies like silicon-based probes, such as Neuropixels [13], which, while revolutionary, are limited

by their rigid construction that leads to tissue damage during chronic implantation. Over time, this rigidity compromises both neuronal function and recording quality.

In contrast, UFTEs are made from ultra-thin, gold-coated polyimide fibers, offering superior flexibility that minimizes tissue damage even during long-term implantation. This allows UFTEs to be implanted deep into the brain while preserving neuronal integrity, enabling stable, months-long tracking of the same neuronal populations with high signal-to-noise ratios.

Additionally, UFTEs enable the simultaneous monitoring of neuronal ensembles across multiple brain regions, facilitating comprehensive mapping of neural circuits. This capability is critical for investigating complex brain functions, such as memory and decision-making, which rely on the coordinated activity of various brain regions. In our lab's study, UFTEs were integrated with a custom Intan-based printed circuit board (PCB), and a 512-channel wireless logger was also developed. This setup allowed us to record neural activity from multiple regions in freely moving animals over several months, providing valuable long-term data.

These properties make UFTEs ideal for brain-machine interfaces and neuroprosthetic systems, where long-term biocompatibility and stable signal quality are critical. Compared to other state-of-the-art technologies like Neuropixels [13], UFTEs provide an optimal combination of spatial resolution, flexibility, and durability, facilitating detailed studies of brain networks and opening possibilities for therapeutic applications, such as the treatment of neurological disorders through precise electrical stimulation.

Figure **??** highlights the UFTE's cellular-scale footprint, designed to minimize tissue damage while enabling implantation into deep brain regions. The figure compares UFTEs to a rigid silicon probe, a flexible planar shank probe, and a neural mesh, showing their relative sizes in relation to nearby neurons' soma and dendrites. Zoomed-in insets depict the effects of different probe geometries on neuronal processes post-insertion.

One challenge in deploying ultra-flexible electrodes is ensuring proper release from the stiff shuttle used for insertion. Premature separation or difficulty in releasing the electrodes can occur if the chemical tether's dissolution rate is not properly calibrated (Figure **??**b). To address this, UFTEs are delivered using a tungsten shuttle, which is retracted post-implantation while the silk coating dissolves, allowing the electrode fibers to remain independently positioned in the brain (Figure **??**c). Remarkably, UFTEs showed stability even 3.5 months post-implantation, as demonstrated by the images of implanted bundles alongside stained neurons and microglia (Figure **??**d).

Despite the development of the 512-channel neurologger, we face significant challenges in expanding the number of recording channels beyond this point. This limitation is not due to the fabrication of the UFTEs themselves, as we are capable of creating bundles with a much higher number of electrode pads or implanting additional bundles. Instead, the bottleneck arises from the interconnection between the flexible electrodes and the external circuitry, specifically the Intan chips used in our current system. Increasing the

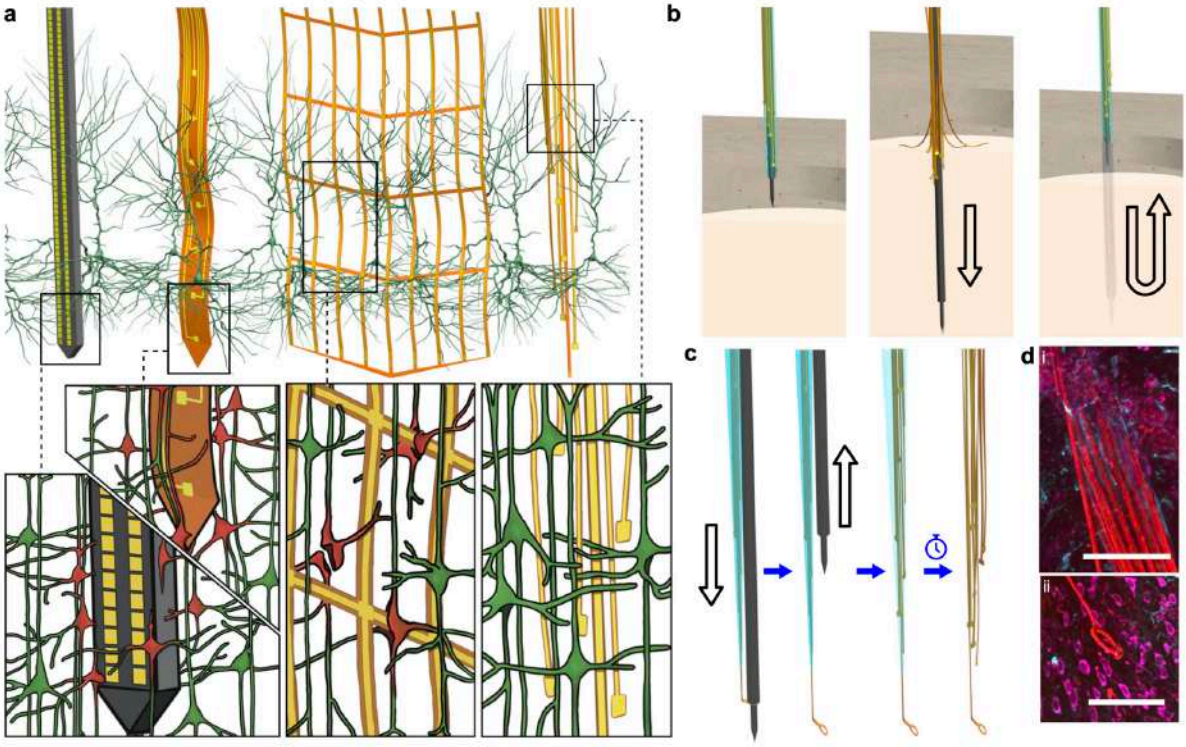


Figure 1.2.: **Ultra-Flexible Tentacle Electrodes (UFTEs)** [1] with cellular-scale footprint reduce tissue damage and are implantable into deep brain regions. **a**: Drawn-to- scale comparison of the geometries of a rigid silicon probe, flexible planar shank probe, neural mesh, and UFTE (from left to right) with reference to the soma and dendrites of surrounding neurons. Zoomed insets (bottom) show the impact of different probe geometries on neuronal processes after insertion into the brain. **b:** Potential failures during the insertion of ultra-flexible electrode arrays coupled to a stiff shuttle purely by chemical tethering (left) **c**: Delivery of the UFTE bundle into the brain, where the bundle isinserted into the brainwith the help of a tungsten shuttle, the shuttle is retracted, and the silk coating is dissolved, leaving the electrode fibers independent from each other. **d:** UFTE 3.5 months post-implantation. (i) UFTE bundle (red) shown with stained neurons (magenta) and the microglia (cyan). (ii) Loop of UFTE bundle in the same brain slice. Scale bars are 100 um.

number of channels would require the use of more Intan chips and stacking additional custom PCBs, which is not an ideal solution due to the added rigidity and volume of the headstage.

To address this issue, we explored the use of CMOS-MEAs, which provide up to 1024 channels in a smaller footprint compared to our existing PCBs. However, the primary challenge in this approach lies in establishing a stable interconnection between

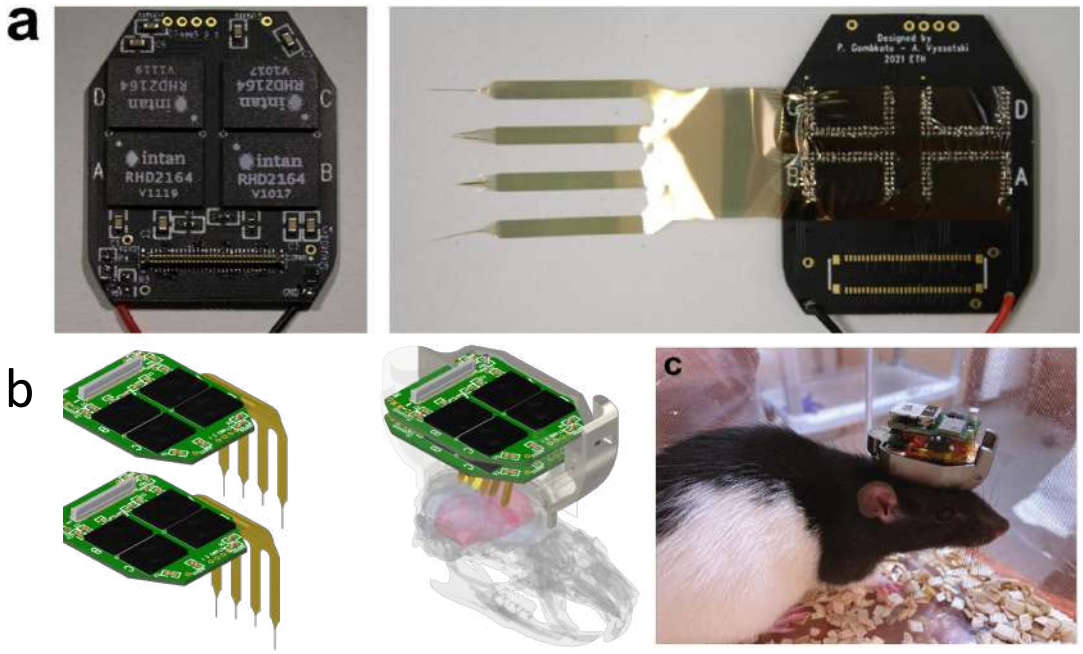


Figure 1.3.: **UFTEs assembly to custom PCBs. A:** The custom 256-channel Intan-based headstage used for chronic recordings. **B:** Stacking of two PCBs, each capable of recording 256 channels, for the wireless 512-channel neurologger. **C:** Implantation of the electrode array in a freely moving rat.

the MEA1k and our flexible electrodes, particularly given the smaller pitch size required for such high-density recordings.

1.3. Earlier Approaches to MEA1k Electrode Assembly

In previous efforts conducted by our lab [14], various assembly approaches were explored to bond flexible electrodes with the CMOS-based MEA1K platform. However, none of these methods fully achieved the projects objectives, primarily due to challenges with precision alignment, low interconnection yield, and insufficient stability of the bonds between the electrodes and the MEA1Ks pixel matrix.

1.3.1. Anisotropic Conductive Film (ACF) Bonding

The project explored the use of anisotropic conductive film (ACF) bonding, a technique in which conductive particles are embedded within a thermosetting resin. When heat and pressure are applied, the particles become conductive, facilitating a bond between the flexible electrodes and the MEA1K. Despite the potential of ACF bonding for high-precision interconnections, the project encountered significant difficulties in aligning the

flexible electrodes with the MEA1K's pixel array. Maintaining consistent pressure during the bonding process proved challenging, and the ACF was easily contaminated after multiple alignment attempts. Consequently, while ACF bonding holds promise for high-density connections, its implementation was limited by the need for precise, one-time bonding and alignment.

Additionally, the process requires the placement of pillars measuring 10-20 um in height on both the MEA1K and electrode sides to ensure proper surface elevation before bonding. This added complexity to the fabrication process, making it significantly more difficult and time-consuming. The fabrication of these pillars introduces another layer of intricacy, which is not practical for highly flexible environments like chronic neural recordings.

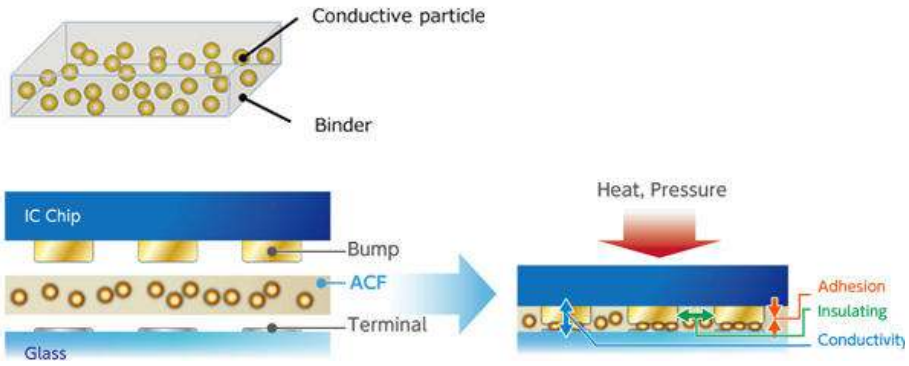


Figure 1.4.: ACF Bondiong Mechanism and Structure

1.3.2. PDMS Microstamping

To avoid the challenges posed by ACF bonding, PDMS microstamping was explored as an alternative approach. This technique aimed to pattern conductive materials onto the MEA1K using a PDMS stamp. However, the first material testedPEDOT:PSSdid not adhere well to the PDMS stamp due to its hydrophobic surface properties, which were incompatible with the hydrophilic nature of PEDOT:PSS. Attempts to modify the PDMS surface using plasma oxidation and PEG coating were unsuccessful, as the hydrophilic properties were lost before effective stamping could occur. The project then explored carbon nanotubes as a potential conductive material, but they did not dissolve properly, rendering them unsuitable for this application. Further tests using gold and silver nanoparticles were suggested, as these materials were expected to bond more effectively with the hydrophobic PDMS surface.

Throughout all the assembly approaches, the precision required for aligning the flexible electrodes with the MEA1K was a significant obstacle. The manual alignment process was unable to consistently achieve the required accuracy to ensure that each electrode was properly connected to the MEA1K pixels without causing cross-talk between adjacent connections. The 17.5 um pixel spacing demanded an extremely high level of precision, which manual alignment could not reliably deliver. To address this, it was proposed that a more sophisticated stamping pattern could be developed to ensure that each electrode aligned with at least one active pixel on the MEA1K, though further testing is necessary to validate this approach.

The findings from this previous project highlighted the difficulties in achieving reliable and scalable bonding for flexible electrodes in high-density neural interfaces like the MEA1K. The key limitations arose from the inability to create stable, low-resistance bonds while maintaining precise alignment. Although ACF bonding and PDMS microstamping showed potential, both methods faced technical challenges, particularly regarding material compatibility and alignment precision. Future work should focus on exploring alternative conductive materials, such as gold or silver nanoparticles, and investigating more advanced microfabrication techniques to achieve consistent bonding and alignment. By overcoming these challenges, the integration of flexible electrodes with high-density CMOS platforms could become feasible for long-term, chronic neural recordings.

Chapter 2

Methods

2.1. UFTEs design considerations

Given the limitations of previously tested approaches, such as ACF bonding and PDMS microstamping, we aimed to explore simpler alternatives that would eliminate the need for additional complex processes, such as placing pillars on top of the electrodes, or the use of specialized equipment like ultrasonic-on-bump bonding [15, 16]. one of the approaches introduced in the literature was Flex2chip design [2]. This interface design2.1, is intended to overcome the key challenge of connecting thousands of electrodes to external circuitry, a problem that hinders current high-channel-count neural interfaces due to the bulkiness of traditional methods like wire bonding and anisotropic conductive films.

We adapted the same principle of flexible connection design and integrated with our UFTEs design and fabrication process. This approach would allow us to bond our flexible electrodes to the MEA1k without the need of any extra fabrication procedures (except for an extra metalization layer). For designing our flexible interconnects, there are several fabrication constraints that we have to respect in order to make sure that the fabrication and the bonding process is successful.

Minimum Distance Between The Flexible Connectors: The distance between the designed flexible connectors must not be less than the pitch size of the pads on the MEA1K side. This is to prevent potential short circuits between the electrodes via the MEA1K pads. Furthermore, even if no short circuit occurs, continuous connections between the rows and columns of MEA1K pads can complicate the identification of the micro-structure pattern, making it difficult to accurately map the MEA1K pads to the electrode bundle placement in the brain. To avoid these complications, the spacing between our flexible interconnects must exceed the pitch size of the MEA1K pads, which is 17.5 um. In the first design of the MEA1K in vivo electrode, a minimum distance

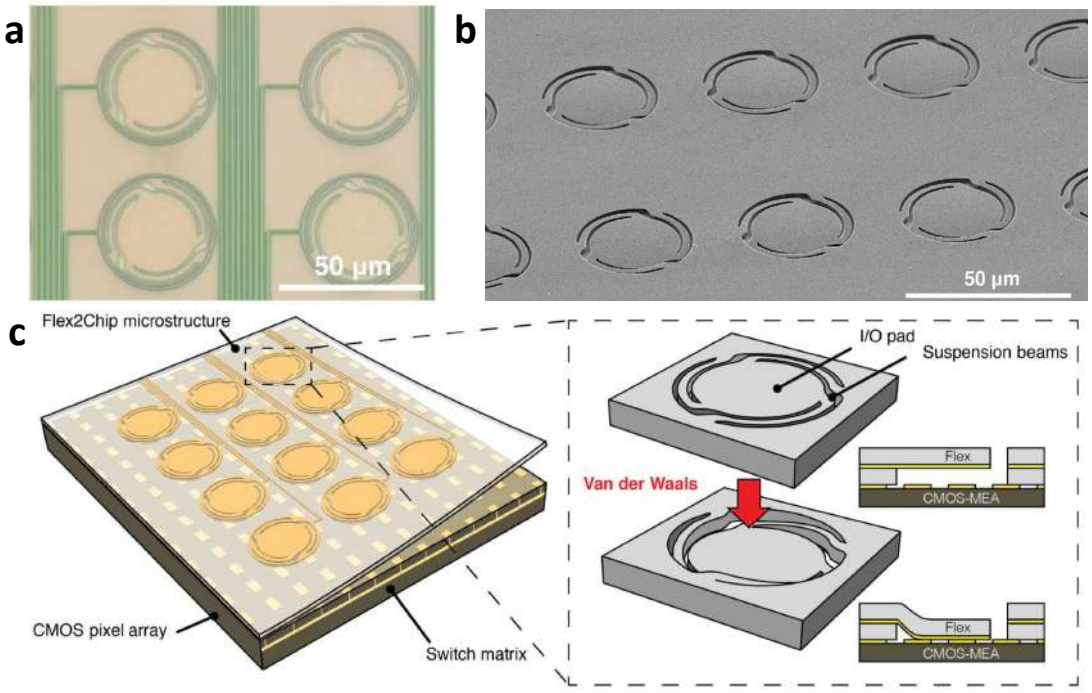


Figure 2.1.: **Flex2chip design[2]**: (**a**) the Flex2chip micro-structure design and routing for the interconnect. (**b**) SEM image of the Flex2Chip device laid on top of a planar glass surface. (**c**) Integration of the Flex2chip design with the underlying CMOS-MEA

of 18 um was used between the flexible interconnects. However, this proximity was too close to the pitch size, leading to potential issues. As a result, in the subsequent design, the distance was increased to 22 um to ensure greater reliability.

Routing Constraints: Each flexible connector placed on the MEA1K side must be individually routed to an electrode recording from the brain. Based on current microfabrication limitations, a minimum wire width of 2 um and at least 3 um spacing between each wire are required to prevent short circuits. These constraints ensure the proper functionality of the electrode array while maintaining the integrity of the routing process.

flex2chip design modification: Although the principle design and working mechanism of our connectors are similar to [2], yet the size of the flexible connectors have been modified in order to make sure that the fabrication process can be reliable. We used three different connector designs, with 60 u, 50 um, and 40 um diameter. The width of the support arms have been kept constant at 2 um and the distance between each of the metal parts is 3 um.

Given the requirement to place at least 1,024 flexible connectors on the MEA1K pad area, which measures 2.1 mm by 3.85 mm, it became clear that designing an interface to fully

utilize the MEA1Ks 1,024-channel capacity was not feasible. We were constrained by both the pitch size of the flexible connectors and the limitations in routing. To overcome these limitations, we implemented a multi-layer metallization design, incorporating two separate metal layers to increase the density of the flexible connectors.

This approach reduced the routing space by 50%, decreasing the wire pitch from 5 um (2 um wire width and 3 um spacing) to an average pitch size of 2.5 um, thanks to the multi-layer structure. As a result, we were able to accommodate 1,116 flexible connectors for a 60 um diameter, 1,164 for a 50 um diameter, and 1,632 for a 40 um diameter. This multi-layer design significantly enhanced the overall routing efficiency while pushing the connector density closer to the required channel count.

After extensive testing with flexible connectors of varying diameters, we selected the design featuring a 50 um diameter and a minimum spacing of 22 um between the pads for our second in vivo electrode design. Additionally, the arrangement and spacing of the flexible connectors were further optimized, allowing us to place 1,278 connectors within the 2.1 mm x 3.85 mm MEA1K pad area. This design achieves an electrode density of 158 electrodes per mmš, which exceeds the maximum recording capacity of the MEA1K by approximately 15%. The additional connectors were included to account for potential connection failures or damage during the assembly process, ensuring the system retains sufficient operational electrodes.

Electrode Bundle Design The flexible connectors are routed toward four electrode bundles, comprising a total of 1,160 electrode channels. This number exceeds the MEA1K capacity by approximately 15%, as we anticipated that some connectors might not attach properly to the MEA1K pads or could become damaged during the assembly process. Therefore, including a margin for error was essential to ensure sufficient functioning electrodes.

In the initial in vivo experiment, two bundles were routed on the top side of the flexible connectors and two were routed downward. This design minimized the width of the flexible connector area, which was advantageous since the MEA1Ks side areas are covered with epoxy. Minimizing the width was preferred to avoid contact between the fabricated electrode device and the epoxy. The total width of the flexible connector portion in this design was 4,342 um, slightly exceeding the 3,850 um width of the MEA1K pad area.

However, having two bundles on each side complicated the surgical implantation, as one side (with two electrode shanks) needed to be folded onto the other side so that all the shanks could be placed on the same side during surgery. Moreover, this structure made it more difficult to lift the device from the silicon wafer and to assemble it onto the MEA1K chip.

To address these issues, the design was modified for the next in vivo surgery, placing all four shanks on the same side. While this revision increased the side width to 5,640 um, complicating the assembly as the edges had to fold over the epoxy, it simplified the surgical process. We attempted to remove as much epoxy as possible using a soldering

iron heated to 310°C from the sides of the MEA1K PCB chip. This allowed us to create a width of approximately 5,800 um between the epoxy on both sides, providing ample space for the flexible device to lie flat on the MEA1K CMOS chip surface. However, this process caused one of the chips to malfunction, likely due to the heat from the soldering iron.

Removing the epoxy requires caution, as any damage to the wire bonds could render the chip unusable. Ideally, future chips should be packaged with minimal epoxy, and this should be communicated to our collaborators in the Bioengineering Laboratory during the wire bonding process to avoid these complications.

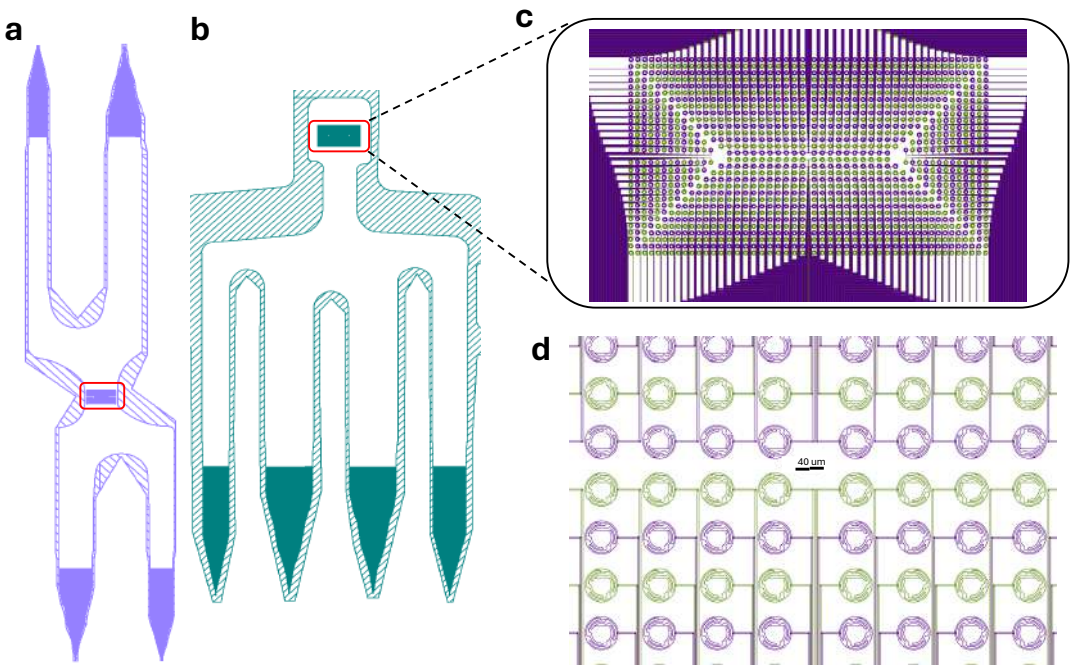


Figure 2.2.: **Overview of the Electrode Design**: (**a**) Electrode design used in the first in vivo experiment with UFTE-MEA assembly. The two shanks are positioned on opposite sides, requiring one side to be bent onto the other, which resulted in excessive tension and complications during the assembly process. (**b**) Electrode design for the second in vivo experiment, with all shanks positioned on the same side, simplifying both assembly and implantation. (**c**) Overview of the Flex2Chip interface, displaying the two metal layers (green and purple). A total of 1,278 flexible contact pads are arranged within a 2.1 mm x 3.85 mm area. (**d**) Close-up view of the flexible interface design used in the second MEA1K experiment, featuring 50 um diameter pads with a minimum spacing of 22 um between the pads.

The design of the electrode shanks ensures that there is one recording electrode per fiber, with each fiber being mechanically independent of the others. This results in a cross-sectional area of 16.8 um^2 for a single fiber, which is approximately one order of magnitude smaller than the cross-sectional area of the soma of a typical neuron.

One key modification in the design of the UFTEs used in this project is the inclusion of two contact pads on each electrode fiber, providing higher spatial resolution. These dual contacts are spaced 20 um, 30 um, or 40 um apart (center-to-center), depending on the specific brain region where they will be implanted. This dual-contact design offers improved precision for clustering single units and facilitates more accurate spike sorting, ultimately enhancing the analysis of neural activity.

2.2. Fabrication Process

The fabrication was carried out at the ETH Zurich clean-room facility (FIRST Micro- and Nanotechnology Center). Although most steps followed the protocol described in the main paper [1], modifications were made to the metal coating, lift-off process, and the etching steps to accommodate the multi-layer design of our electrodes.

The electrode consists of two Ti/Au metal layers sandwiched between three polyimide layers. Polyimide, a high-performance polymer widely used in microelectronic and biomedical applications, is particularly suitable for neural implants due to its thermal, chemical, and mechanical stability, as well as its biocompatibility [17, 18, 19]. Polyimides robustness and flexibility make it an ideal material for long-term, flexible electrode arrays used in neuroscience research.

Base Polyimide Coating: The fabrication begins with a base polyimide coating on a 4-inch silicon wafer. A vial of PI2610 (HD Microsystems) is removed from the deep freezer and allowed to reach room temperature for at least two hours. Approximately 4 ml of PI2610 is then dispensed using a 3 ml Pasteur pipette, ensuring that any air bubbles are burst with a clean 1 ml pipette. The wafer is spin-coated using the following parameters: 500 rpm for 1 seconds with a 2 second acceleration, followed by 5000 rpm for 40 seconds with a 4 second acceleration , yielding a 1.2 um thick polyimide layer. The wafer is soft-baked on a hotplate at 100°C for 200 seconds, then transferred to a glass petri dish and cured in a programmable oven (CLO-2AH-S, Koyo ThermoSystems) at 300°C, using program 4.

Patterning of the photo-resist for the first metal layer coating: in the previous fabrication steps used for UFTEs, the ma-N-1420 (Micro Resist Technology GmbH) was used for the metal patterning. However, after our first attempt to fabricate the In Vivo electrodes, we noticied that this photoresist was not able to achieve a stable pattern with the resolution required for the structures in our design. Specifically the long 2 um wide wires with 3 um spacing were unstable, due to undercut of the resist after the development. Therefore you could see the photo-resist detaching from the wafer during the development. therefore, we had to switch to another photo-resist with a higher resolution that can support our pitch sizes. We chose the AZ nLOF 2020 (MicroChemicals GmbH) Photoresist, this is a type of negative photoresist used in the photolithography process for microfabrication. It is designed for applications where

high-resolution patterning is required, especially for devices like MEMS (Micro-Electro-Mechanical Systems), microfluidics, and advanced packaging technologies. The "nLOF" stands for negative Lift-Off, which means it is optimized for lift-off processes where the patterned resist is removed after deposition, leaving behind structures such as metals on the substrate. The AZ nLOF 2020 can achieve fine patterns with critical dimensions as small as 0.5 um, making it suitable for high-precision devices such as our case.

We first dispense the AZ nLOF 2020 photoresist onto the polyimide-coated wafer and spin-coat it using a two-step process: initially at 500 rpm for 10 seconds with a 2.5-second acceleration, followed by 3000 rpm for 40 seconds with a 2-second acceleration. The wafer is then soft-baked at 110°C for 60 seconds. Using the *EVG 620 NT mask alignment technology*, the photoresist is patterned via UV exposure (UV mix 20% at 180 mJ/cm^2) through the first metal layer mask.

A post-exposure bake (PEB) is performed at 110°C for 60 seconds to further enhance the cross-linking of the resist. The photoresist is then developed with *AZ 726 MIF developer* (MicroChemicals GmbH) for 60 seconds, followed by rinsing under deionized water (dH_2O) for 60 seconds. Finally, the quality of the developed pattern is inspected under an optical microscope (*Nikon Eclipse L200D*) to ensure the accuracy and resolution of the process.

First Metal Coating and Lift-off: We first deposited a 10 nm titanium and 150 nm gold layer using electron beam evaporation (*Plassys MEB550S*), with Ar plasma etching beforehand to improve the adhesion of the metal layer to the polyimide and clean the surface. Electron beam evaporation is a physical vapor deposition (PVD) technique that uses an electron beam to vaporize the source material, which then condenses onto the wafer surface to form a thin film. This process is performed in a vacuum chamber, allowing for high-purity, uniform films. In our case, the 10 nm titanium serves as an adhesion layer, while the 150 nm gold provides the conductive layer. The versatility of electron beam evaporation ensures precise control over the thickness of the deposited layers, making it ideal for microelectronic applications like ours.

After the metal deposition, the wafer was placed in a beaker filled with acetone, with a lid to prevent evaporation. It was left at the wet bench for one hour, followed by lift-off through air-blown agitation using a pipette.

For narrow and enclosed areas, such as the flexible connectors, it is important to perform more aggressive air-blown agitation due to the complexity of the metal traces, which can make the lift-off process more difficult. The wafer was then left in the acetone bath for an additional 3-4 hours. Following this, the wafer was sonicated at level 3 (132 kHz ultrasound) for 20 minutes and subjected to air-blown agitation using a 1 mL pipette in two consecutive acetone baths.

The lift-off process was monitored under a binocular microscope while the wafer remained in the acetone bath to ensure the quality of the lift-off. Special attention was given to the wire traces to confirm that no residual metal particles had attached to them. Afterward,

the wafer was placed in an isopropanol bath for 45 seconds, rinsed under deionized water (dH_2O) for 60 seconds, and then dried with nitrogen. The quality of the metal patterning was subsequently inspected under an optical microscope.

Insulating polyimide coating: Afterward, the polyimide base coating is reapplied as described earlier. A vial of PI2610 (HD Microsystems) is taken from the deep freezer and allowed to reach room temperature for at least two hours. Approximately 4 ml of PI2610 is then dispensed using a 3 ml Pasteur pipette, ensuring that any air bubbles are removed with a clean 1 ml pipette. The wafer is spin-coated with the following parameters: 500 rpm for 1 second with a 2-second acceleration, followed by 5000 rpm for 40 seconds with a 4-second acceleration. This results in a 1.2 um thick polyimide layer. The wafer is then soft-baked on a hotplate at 100°C for 200 seconds before being transferred to a glass petri dish and cured in a programmable oven (CLO-2AH-S, Koyo ThermoSystems) at 300°C using program 4.

Patterning of the photo-resist for the second metal coating: We dispense the AZ nLOF 2020 photoresist onto the polyimide-coated wafer and spin-coat it using a two-step process: initially at 500 rpm for 10 seconds with a 2.5-second acceleration, followed by 3000 rpm for 40 seconds with a 2-second acceleration. The wafer is then soft-baked at 110°C for 60 seconds. Using the *EVG 620 NT mask alignment technology*, the photoresist is patterned via UV exposure (UV mix 20% at 190 mJ/cm^2) through the second metal layer mask. the exposure rate is slightly higher than the previous metal layer as the thickness of the polyimide base has increased and thus the reflection of the UV light is more absurbed by the polyimide. it's noteworthy that the alignment should be as well as possible between these two layers as a mismatch can cause problems in the further steps.

A post-exposure bake (PEB) is performed at 110°C for 60 seconds to further enhance the cross-linking of the resist. The photoresist is then developed with *AZ 726 MIF developer* (MicroChemicals GmbH) for 60 seconds, followed by rinsing under deionized water (dH_2O) for 60 seconds. Finally, the quality of the developed pattern is inspected under an optical microscope (*Nikon Eclipse L200D*) to ensure the accuracy and resolution of the process.

Second Metal Coating and Lift-off: we then move again to the metal coating and lift-off step for our second metal layer, we use the same method as previously described. We first deposited a 10 nm titanium and 150 nm gold layer using electron beam evaporation (*Plassys MEB550S*), with Ar plasma etching beforehand to improve the adhesion of the metal layer to the polyimide and clean the surface.

After the metal deposition, the wafer was placed in a beaker filled with acetone, with a lid to prevent evaporation. It was left at the wet bench for one hour, followed by lift-off through air-blown agitation using a pipette.

For narrow and enclosed areas, such as the flexible connectors, it is important to perform more aggressive air-blown agitation due to the complexity of the metal traces, which can

make the lift-off process more difficult. The wafer was then left in the acetone bath for an additional 3-4 hours. Following this, the wafer was sonicated at level 3 (132 kHz ultrasound) for 20 minutes and subjected to air-blown agitation using a 1 mL pipette in two consecutive acetone baths.

The lift-off process was monitored under a binocular microscope while the wafer remained in the acetone bath to ensure the quality of the lift-off. Special attention was given to the wire traces to confirm that no residual metal particles had attached to them. Afterward, the wafer was placed in an isopropanol bath for 45 seconds, rinsed under deionized water (dH_2O) for 60 seconds, and then dried with nitrogen. The quality of the metal patterning was subsequently inspected under an optical microscope.

Insulating polyimide coating: Afterward, the polyimide coating step is reapplied as described earlier. A vial of PI2610 (HD Microsystems) is taken from the deep freezer and allowed to reach room temperature for at least two hours. Approximately 4 ml of PI2610 is then dispensed using a 3 ml Pasteur pipette, ensuring that any air bubbles are removed with a clean 1 ml pipette. The wafer is spin-coated with the following parameters: 500 rpm for 1 second with a 2-second acceleration, followed by 5000 rpm for 40 seconds with a 4-second acceleration. This results in a 1.2 um thick polyimide layer. The wafer is then soft-baked on a hotplate at 100°C for 200 seconds before being transferred to a glass petri dish and cured in a programmable oven (CLO-2AH-S, Koyo ThermoSystems) at 300°C using program 4.

Patterning of the Photoresist Mask for Reactive Ion Etching (RIE): This step is crucial for removing the polyimide between the electrodes to form separate fibers, outlining the borders of the device, and removing the polyimide from the top of the flexible connectors to enable the gold surface of the connectors to attach to the MEA1K surface. The process begins by dispensing *AZP4620 photoresist* (MicroChemicals GmbH) onto the wafer, using the following spin-coating parameters: 500 rpm for 10 seconds with a 2.5-second acceleration, followed by 4000 rpm for 40 seconds with a 4-second acceleration. The wafer is then soft-baked at 110°C for 180 seconds on a hotplate.

Using the *EVG 620 NT mask alignment system*, the photoresist is patterned via UV exposure at 430 mJ/cm^2 through the etching mask. After exposure, the wafer is developed in *AZ726* (MicroChemicals GmbH) for 5 minutes, rinsed under running deionized water (dH_2O) for 40-45 seconds, and dried with nitrogen blow.

Reactive Ion Etching (RIE): The wafer with the patterned *AZP4620 photoresist* is placed into the RIE80 system (Plasmalab 80 Plus, Oxford Instruments) to perform Reactive Ion Etching.

Reactive Ion Etching (RIE) is a dry etching technique that combines physical ion bombardment and chemical reactions to precisely remove material from a substrate. In this process, a plasma is generated from gases such as O_2 and CF_4, which are ionized and directed towards the substrate. For polyimide etching, O_2 is used as the primary reactive gas, breaking down the polymer into volatile compounds like CO and CO_2, while CF_4

enhances etching selectivity and rate. This enables anisotropic etching, where vertical etching dominates, reducing lateral undercutting and producing well-defined features. RIE offers excellent control over etching depth and patterning, making it ideal for microfabrication applications such as MEMS, integrated circuits, and flexible electronic devices employing polyimide layers.

Given the multi-layer design of our electrodes, this etching step requires adjustment to etch through 3 um of polyimide, compared to the 2 um in the previous UFTEs fabrication process. In the RIE80 system, we etch using 20 sccm CF_4, 100 sccm O_2, at a pressure of 100 mTorr and a power of 90 W. The etching process is carried out in 2 cycles of 6 minutes (the exact duration can vary slightly depending on the condition of the machine), followed by a final cycle of 6.5 minutes. After each cycle, we inspect the wafer under an optical microscope to ensure that the etching is complete. If the etching appears satisfactory, the wafer is immersed in an acetone bath to dissolve the excess photoresist. It is then placed in an isopropanol bath for 45 seconds, rinsed under running deionized water (dH_2O) for 40-45 seconds, and dried with a nitrogen blow. Finally, we inspect the wafer under the microscope to assess the quality of the etching.

Sacrificial Parylene Coating:

The sacrificial parylene coating process is widely used in microfabrication to protect delicate layers such as PEDOT:PSS during subsequent processing steps. Parylene is a biocompatible, chemically resistant polymer applied in a vapor phase, forming a uniform, conformal coating over complex structures. This sacrificial layer shields sensitive materials from potential damage caused by etching, handling, or other fabrication processes that might compromise their functionality or electrical properties.

In our process, the sacrificial parylene layer is used specifically for patterning PEDOT:PSS. It protects the conductive polymer throughout the remaining fabrication steps, and once these steps are completed, the parylene is removed. This approach ensures the integrity and functionality of the PEDOT:PSS layer while enabling the precision required for fabricating neural electrodes and other flexible devices.

Before applying the sacrificial parylene coating, the wafer is spin-coated with **Micro90**. Micro90 is a surfactant that creates a thin, uniform layer on the wafer surface, facilitating the subsequent removal of the parylene coating. By applying a 2% solution of Micro90 (International Products Corporation) before the parylene deposition, we ensure that the parylene layer can be easily removed by immersing the wafer in water, as Micro90 reduces the adhesion between the parylene and the underlying surface. The wafer is spin-coated with Micro90 using the spinner in the Neurotechnology lab with the following parameters: 1000 rpm for 60 seconds with a 5-second acceleration time. Due to its low viscosity, Micro90 uniformly covers the entire wafer before spinning.

After spin-coating with Micro90, a 4.5 um layer of sacrificial parylene C is deposited on the wafer using chemical vapor deposition (PDS 2010 Labcoter 2; Specialty Coating Systems Inc.). This sacrificial parylene layer is crucial for patterning PEDOT:PSS via a

dry lift-off process. After the necessary processing steps, the wafer is immersed in water, where the Micro90 layer facilitates the easy dissolution and removal of the parylene coating. This method ensures that the underlying PEDOT:PSS layer remains intact and functional, while providing the precision needed for the fabrication process.

The use of Micro90 is essential not only for simplifying the parylene removal but also for preventing the use of harsh chemicals or aggressive etching processes, which could damage the delicate PEDOT:PSS layer.

Patterning of the Photoresist Mask for Reactive Ion Etching (RIE): This step is for removing the polyimide and the parylene layer from on top of the contact pads of the electrodes and exposing the gold surface of the contact pads to later be coated with PEDOT:PSS. The process begins by dispensing *AZP4620 photoresist* (MicroChemicals GmbH) onto the wafer, using the following spin-coating parameters: 500 rpm for 10 seconds with a 2.5-second acceleration, followed by 4000 rpm for 40 seconds with a 4-second acceleration. The wafer is then soft-baked at 110°C for 180 seconds on a hotplate.

Using the *EVG 620 NT mask alignment system*, the photoresist is patterned via UV exposure at 430 mJ/cm^2 through the electrodes mask. After exposure, the wafer is developed in *AZ726* (MicroChemicals GmbH) for 5 minutes, rinsed under running deionized water (dH_2O) for 40-45 seconds, and dried with nitrogen blow.

Reactive Ion Etching (RIE):
The wafer with the patterned *AZP4620 photoresist* is placed into the RIE80 system (Plasmalab 80 Plus, Oxford Instruments) to perform Reactive Ion Etching (RIE).

Due to the multi-layer design of our electrodes, the etching process must be adjusted to etch through 2 um of polyimide for the deeper contact pads, along with an additional 4.5 um thick parylene layer, compared to the 1 um polyimide layer used in previous UFTEs fabrication. In the RIE80 system, the etching is performed using 20 sccm CF_4 and 100 sccm O_2 at a pressure of 100 mTorr and a power of 90 W. The process is carried out in four cycles of 5 minutes each (with slight variations depending on machine conditions), followed by a final 6-minute cycle to etch through the extra 1 um polyimide layer. After each cycle, the wafer is inspected under an optical microscope to ensure complete etching.

In a previous attempt, I noticed that the photoresist layer was not thick enough to protect other parts from the extended etching, as an additional 6 minutes were required. Due to time constraints for the surgery, I was unable to optimize the photoresist patterning parameters and instead performed the etching in two stepsfirst removing the polyimide from the contact pads, then performing the sacrificial parylene coating and etching it again. For future fabrication processes, I recommend increasing the thickness of the *AZP4620 photoresist* by lowering the spin-coating rate, enabling the parylene-polyimide etching to be completed in one step. This adjustment will also improve the alignment between the two layers, enhancing the quality of the PEDOT:PSS coating.

After the etching cycles are complete and the quality is confirmed, we use AZ 100 Remover in an 80°C water bath instead of acetone to remove the photoresist. AZ 100 Remover is specifically formulated to dissolve photoresist without being aggressive toward other layers, such as parylene. Unlike acetone, which might prematurely lift off the parylene layer (due to the presence of Micro90), AZ 100 Remover selectively dissolves only the photoresist, leaving the sacrificial parylene intact until its designated removal step, which involves immersion in water.

The wafer is immersed in the hot AZ 100 bath for 15 minutes, with gentle agitation using a 3 mL pipette every 5 minutes. As the photoresist dissolves, the AZ 100 solution turns purple. Afterward, the wafer is transferred to a second, clean AZ 100 bath for another 15 minutes, this time without agitation. By this point, the photoresist should have been fully removed, and the color of the solution should not change.

Following the AZ 100 bath, the wafer is rinsed in an isopropanol bath for 45 seconds but not with water, as this would cause the parylene layer to come off. The wafer is then dried with a nitrogen blow, but the primary drying is done with a 2-minute bake on a 100°C hotplate. Finally, the quality of the etching is checked under an optical microscope to ensure the process was successful.

PEDOT:PSS Coating:
Poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) is widely used in neural electrode applications due to its unique combination of electrical conductivity, flexibility, and biocompatibility [20, 21, 22]. One of its primary advantages is its high electrical conductivity, which enables efficient signal transduction between neural tissue and electronic devices. PEDOT:PSS has a lower impedance compared to traditional metal electrodes like gold or platinum, significantly improving the signal quality by reducing noise and increasing the electrode's sensitivity to small electrical signals. This is particularly crucial in neural recording and stimulation applications, where high signal fidelity is needed for precise measurements [20].

Moreover, PEDOT:PSS is flexible and can conform to soft biological tissues, minimizing the mechanical mismatch between rigid electrodes and the surrounding neural tissue [22]. This helps reduce inflammation and tissue damage during long-term implantation. Its biocompatibility makes it an ideal candidate for chronic neural interfaces, where maintaining long-term performance without inducing adverse biological responses is critical [21]. Additionally, PEDOT:PSS can be easily patterned on flexible substrates and integrated into microfabrication processes, further enhancing its appeal in the fabrication of next-generation neural electrodes [20].

We sonicate a mixture of PEDOT:PSS (Clevios PH 1000, Heraeus Epurio), ethylene glycol (Sigma Aldrich), dodecyl benzene sulfonic acid (Sigma Aldrich), and (3-Glycidyloxypropyl) trimethoxysilane (Sigma Aldrich) in an ultrasound bath for 20 minutes. Then, 5 mL of the mixture is dispensed onto the wafer to almost completely cover the surface, particularly focusing on the electrode contact pads. The wafer is spin-coated using the following parameters: 650 rpm for 60 seconds with a 5-second acceleration. After spin-coating,

the wafer is placed in an oven at 140°C for one hour. These steps are repeated two more times to achieve a total thickness of 450 nm.

After allowing the wafer to cool down to room temperature, it is placed in a large beaker of MilliQ water. The sacrificial parylene layer is then gently peeled off using tweezers. Large chunks should come off easily, though extra care is required for smaller pieces that may stick to the solder pads. Caution is necessary to avoid damaging the flexible contact pads, which can be delicate and sensitive to the metal tweezer. To prevent damage, it is advised to remove the smaller pieces by slowly guiding the larger scratched chunks toward their direction.

Upon completing these steps, the fabrication of the flexible devices is finalized.

2.3. Gold-Plating for MEA1k

In the CMOS MEA platform, the platinum microelectrodes are embedded within a 1.6 um passivation stack consisting of alternating SiO_2 and Si_3N_4 layers. This passivation protects the underlying circuits but leaves the contact pads slightly recessed after the reactive-ion etching (RIE) process. To ensure reliable mechanical and electrical contact between the pads and our flex2chip connectors, gold plating is applied to elevate the surface of the MEA1k pads by 3 um1.6 um to reach the top surface and an additional 1.4 um to elevate the contact pads with gold pillars. This ensures that the flexible connectors can physically access and connect to the exposed pads, establishing robust electrical connections.

The gold-plating process was performed using a cyanide-based gold solution ("24K pure gold solution," Gold Plating Services), which provides less porous plating compared to non-toxic alternatives. In our case, a lower current than the recommended value was applied to control the gold thickness. The goal was to achieve a 3 um thick gold layer, which is sufficient to surpass the 1.6 um passivation layer while avoiding short circuits between pads (which are spaced 8.2 um apart). The gold solution was heated to a temperature between 62°C and 71°C, carefully controlled by a closed-loop temperature controller using an IR sensor for precise, contactless measurement. Platinum counter-electrodes were centrally placed at a significant distance from the chip to ensure even plating, while a plastic cup sealed with DC-744 sealant (Dow Corning) was used to prevent leaks.

The recommended current for the MEA1K, with a total contact area of 1.3381 mmš, was approximately 31 uA/mmš. However, a current of 20 uA was applied to plate 3 um of gold. The potential of the pads was set to -1.15 V relative to the reference ground to prevent unintentional gold-plating on the ring electrodes. The process took around 30 minutes, with each 1 um layer requiring approximately 10 minutes of plating time. Maintaining the temperature between 62°C and 71°C was crucial, as deviations could melt adhesives or sealants, potentially contaminating the gold solution and damaging

the electrode pads. Additionally, controlling the electrode potential was essential to avoid unintended gold-plating on the reference electrodes. The pH of the solution was carefully regulated to ensure stable and consistent gold deposition, critical for achieving the desired thickness and quality of the gold layer.

After the plating process, the electrodes were visually inspected under a microscope to ensure uniformity and the correct thickness. Uneven or excessive plating could lead to short circuits, especially given the narrow 8.2 um gap between the pads. Any remaining solution or contaminants were carefully removed to avoid interference with electrode performance.

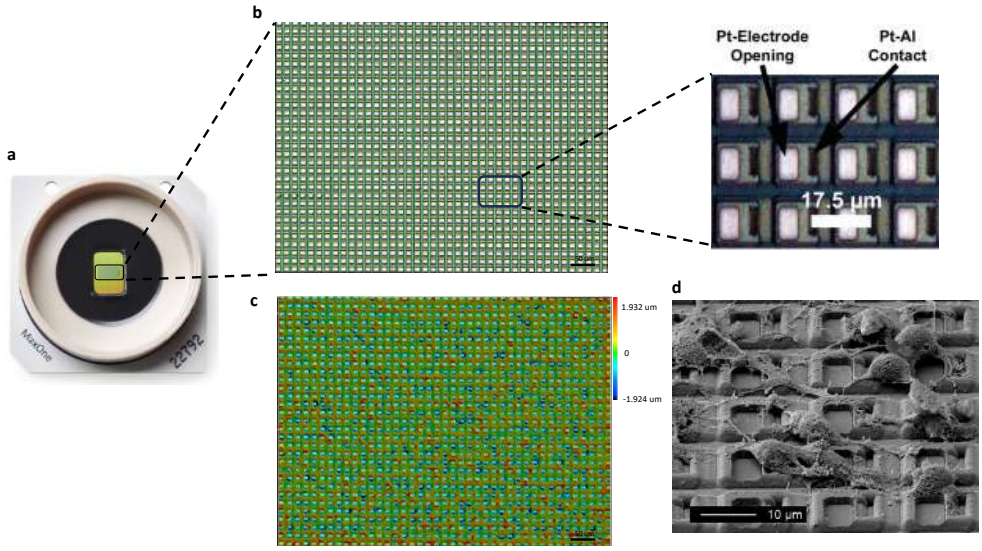


Figure 2.3.: **Flex2chip design[2]**: (**a**) The MEA1k chip from Maxwell Biosystems. (**b**) A close-up view of the electrode pads, showing the Pt-electrode openings. (**c**) A surface profile of the MEA1k, illustrating the recessed recording pads. (**d**) SEM image of the electrode array, displaying a detailed close-up of the microelectrodes after in-house post-processing, where the larger indented platinum rectangles are visible with cultured rat cortical neurons and glial cells growing on the array. [3]

Several challenges were encountered during the gold-plating process. If the gold layer becomes too thick, especially between pads with narrow gaps, it can lead to electrical short-circuits. Maintaining the purity of the gold solution was also critical, as contaminants could affect the deposition rate and the uniformity of the plating. Furthermore,

temperature fluctuations in the solution could compromise the adhesion and overall quality of the gold layer. By carefully controlling factors such as current, temperature, duration, and pad potential, high-quality, robust electrode surfaces were achieved, ensuring long-term functionality for neural recording applications.

2.4. characterization of the UFTE-MEA connectivity

Following the bonding of the flexible contact pads to the MEA1k and securing the assembly with a custom titanium holder, a comprehensive connectivity analysis was performed. The shanks were submerged in saline, and a 5 mV sinusoidal signal at 1 kHz was applied. Each of the 26,400 MEA1k pads recorded for 1 second, and the recorded data was filtered between 995 Hz and 1005 Hz. The peak power within this range was normalized based on the input signal to quantify the fraction of the signal picked up by each pad. The resulting values, ranging from 0 (no signal) to 1 (full 5 mV signal), represented the connectivity measurement for each pad. By arranging the channels according to their physical placement on the MEA1k, we identified the same pattern as the flexible contact pads, allowing us to map the connectivity pattern to the corresponding shanks and trace each MEA1k pad to its electrode placement in the brain.

Additionally, the phase of the input signal was compared to the recorded phase to assess whether the connections were ohmic or capacitive. A phase shift of 0 degrees indicated a fully ohmic connection, while a 90-degree shift demonstrated a capacitive connection.

For impedance measurements, the shanks were inserted into the saline, and a sinusoidal current with 2 um amplitude was applied using an external current generator (STG4000, Warner Instruments). Connectivity measurements were used to determine which pads had a high signal-to-noise ratio, and only these were measured for impedance while all other channels were set to off to prevent interference.

To verify the long-term stability of the connections, we assembled the UFTEs onto an MEA1k chip and monitored connectivity values over three weeks. The assembly was kept at room temperature under standard conditions, and connectivity maps were recorded on nine separate days throughout the duration. This ensured that the connections remained stable over time, as required for experiments spanning weeks to months. At the conclusion of the tests, the electrodes were disassembled for further analysis.

2.5. Surgical assembly for In-vivo implantation

The surgical procedure is the same as [1]. Following the bonding process of the flexible contact pads to the MEA1k and securing the assembly with the costume titanium holder each electrode bundle was then dipped in a solution of 0.2 g/mL PEG4000, and coated three times with a silk fibroin solution, ensuring the loops at the electrode tips remained unclogged.

Tungsten shuttles were prepared by cutting 15 mm segments from 50 um diameter tungsten wires, and partially inserting them into glass capillaries to facilitate handling. These wires were trimmed to the desired length and their tips were thinned to a 20 um diameter using electrochemical etching in 0.9 M KOH. The tip of each wire was then sharpened by applying 2V DC while submerging the tip in KOH, followed by rinsing with deionized water. Tip dimensions and sharpness were confirmed under a microscope.

Once the UFTEs were soldered to the headstage and the tungsten shuttles were prepared, the headstage was mounted to a 3D-printed electrode holder attached to a stereotaxic arm. An opposing stereotaxic arm held a tungsten shuttle mounted via a 3D-printed pipette holder. Using stereotaxic positioning, the tip of the shuttle was inserted into the loop of the most anterior electrode bundle, and the bundle was secured to the shuttle's glass pipette with a PEG4000 solution for additional stability. The shuttle was then transferred to its respective slot in the electrode holder, and this process was repeated for the remaining electrode bundles.

For the TitaniumHelmet and surgical procedures, we used a titanium-based cap that is fully biocompatible and designed for secure and stable long-term recording. The helmet, consisting of a base, left and right enclosures, and a top cover, was secured to the rat's skull using titanium screws, while the electrode bundles were implanted in a predefined anterior-to-posterior sequence. Once implanted, the electrode sites were covered with a silicon elastomer before attaching the headstage to the helmet. After closing the helmet, we cleaned the surgical site and sutured any remaining gaps in the scalp, allowing the rat to recover in a clean, warm cage with food and nesting material available.

We implanted four UFTE bundles targeting six distinct brain regions: the medial prefrontal cortex (mPFC), specifically covering the infralimbic (IL), prelimbic (PrL), and cingulate (Cg1) areas, along with the retrosplenial cortex (RSC), the dorsal hippocampus (dHPC), and the intermediate hippocampus (iHPC). Each bundles tungsten shuttle was first transferred from a custom 3D-printed electrode holder to a stereotaxic arm, which then guided the insertion into the brain at a controlled speed of 12.5 um/s. Thanks to the flexibility of the polyimide cable that connects the electrode bundle to the headstage, it was possible to insert bundles at various angles, allowing for precise placement in diverse brain regions. In total, 2 electrode bundles were successfully implanted into these targeted regions across four rats.

The mechanical tethering between the electrode bundles and the tungsten shuttles allowed for smooth penetration through the dura mater with minimal disruption to the protective meninges.

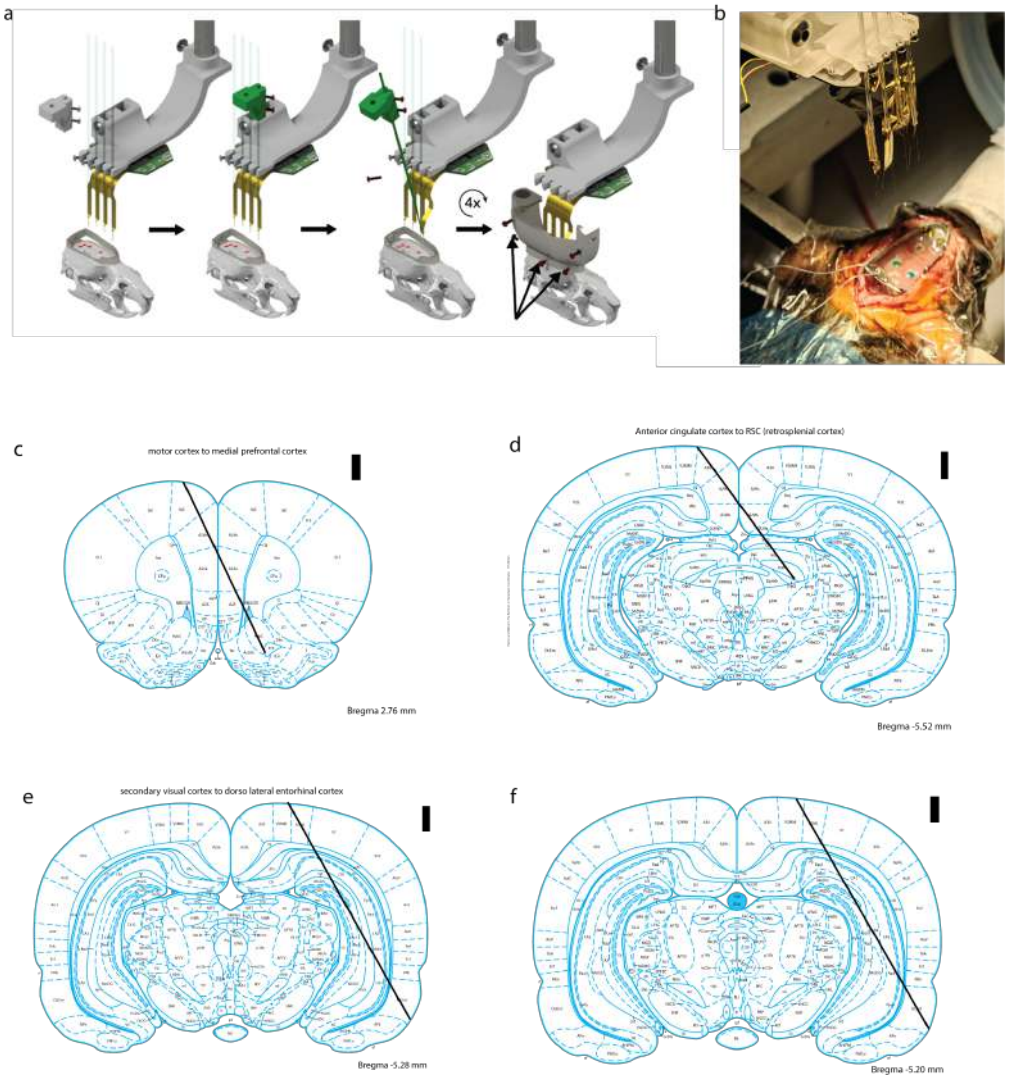


Figure 2.4.: **Surgical Implantation Procedure (a)** Steps involved in the surgical implantation process for each electrode bundle. The components moved during each stage are highlighted in green, while screws inserted at various steps are indicated in red along with arrows for clarity. **(b)** The precise implantation of the electrode bundles into the target brain regions. **(c)** A 250-channel shank, covering from the motor cortex to the medial prefrontal cortex (mPFC), which includes the infralimbic and prelimbic cortices. **(d)** A 200-channel shank spanning the anterior cingulate cortex (ACC) to the retrosplenial cortex (RSC). **(e)** A 250-channel shank traversing from the anterior cingulate cortex to the retrosplenial cortex (RSC). **(f)** A 354-channel shank positioned to cover the secondary visual cortex (V2) and extend into the dorsolateral entorhinal cortex. **(g)** A 356-channel shank similarly spanning from the secondary visual cortex to the dorsolateral entorhinal cortex.

2.6. High-Density Wireless Neural Recording System

Investigating brain mechanisms underlying natural behaviors necessitates simultaneous recording of neural activity across multiple brain areas in unrestrained, highly mobile animals. The multichannel electrodes developed in this thesis provide an optimal solution for capturing neural activity due to their long-term stability and high channel density. However, the acquisition of data from such electrodes presents a significant challenge. A detailed analysis of brain activity requires continuous recordings of neural potentials across many channels. To achieve reliable spike sorting, the minimal sampling rate should be at least 20 kHz. Consequently, a recording system for 1,024 channels with a 10-bit resolution must support a continuous data acquisition rate of approximately 25.6 MBps, with the capacity to record for at least one hour. Additionally, the total weight of the system, including the head implant, protective casing, electrodes, and battery, must not exceed 15 g to allow for mounting on small laboratory animals, such as rats or marmoset monkeys. Since existing recording systems with such channel capacities and form factors are not available, we designed and developed a custom neural logger based on our MEA-UFTE configuration.

The designed neural logger is illustrated in 2.5. It is equipped with an onboard microSD memory card connector, which allows data to be stored locally. Optional wireless data transmission is supported via the Murata Type 1LV WiFi/BT module located near the microSD connector. To complement neural data with behavioral and environmental context, the logger is equipped with motion and environmental sensors. These include a 3D accelerometer, 3D gyroscope (LSM6DSOX), and 3D magnetometer (IIS2MDC) for comprehensive inertial tracking. Environmental sensors include a digital microphone (SPH0655LM4H-1), an illumination sensor (TSL25723), and a pressure sensor (BMP388). The microphone is particularly useful for studying vocalizations during social interactions, and additional microphones can be attached via auxiliary connectors. The high-accuracy pressure sensor can serve as an altimeter, useful for monitoring altitude changes in experiments involving flying animals. Precise synchronization with external experimental equipment, such as video cameras, can be achieved through an infrared (IR) link, using the attached TSOP37338 IR sensor or a 3D IR tracking module similar to those used in virtual reality systems.

At the core of this recording system is the i.MX RT600 Crossover microcontroller (MCU), featuring Arm Cortex-M33 and DSP cores in a 249-ball, 0.4 mm pitch FOWLP package (NXP Semiconductors). This MCU was selected for its balance of low power consumption and high performance, capable of clock speeds up to 300 MHz (M33 core, 1.13V core voltage) due to its 28-nm technology. The MCU is equipped with 4.5 MB of RAM for data caching before storing it on the microSD card. A key advantage of this MCU is its fast peripheral interfaces, typically found in larger processors. These include two SD card host controllers (SDHC), which are certified for clock rates up to 208 MHz but have been tested successfully at 250 MHz in our application. The SDHCs provide a sustained

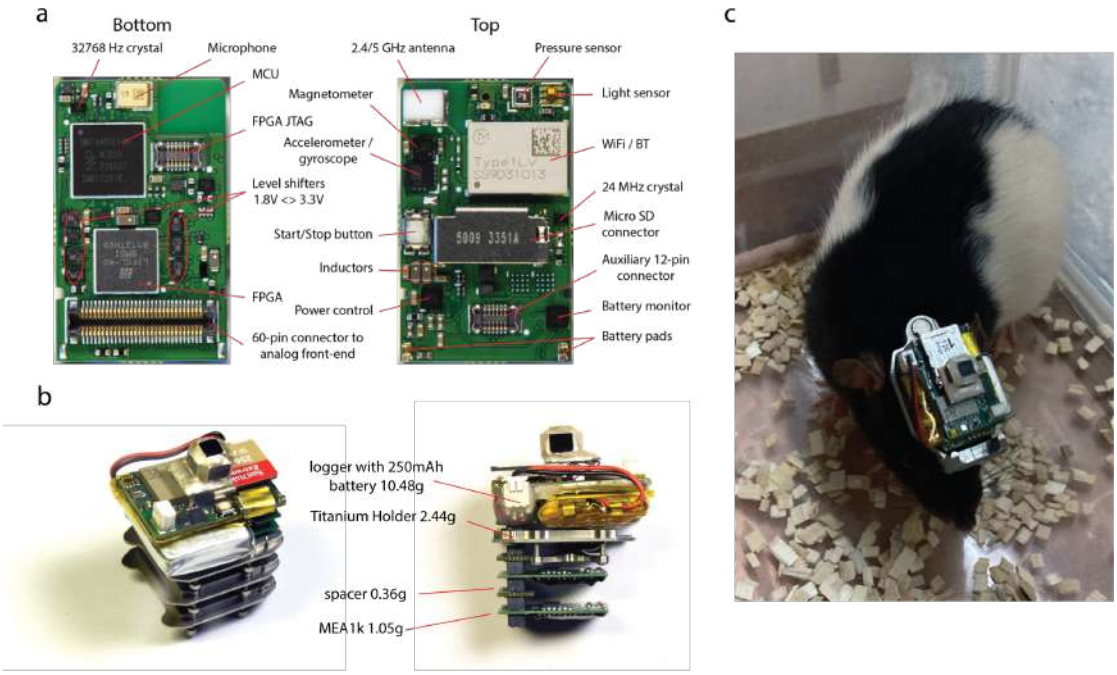


Figure 2.5.: **High-density wireless neural recording system.**: (**a**) The size of the board is 28 mm by 17 mm, with a total weight of 3.25 g when the microSD card is included. The logger is highly versatile, with a 60-pin connector that enables attachment of various neural headstages: up to five stackable 256-channel headstages based on Intan chips (providing up to 1,280 channels in total) or up to three stackable 1,024-channel headstages based on MEA1k chips (supporting a total of 3,072 channels). (**b**) Logger board assembled with three MEA1k boards, enabling 3,072 channels. (**c**) Implanted wireless logger board with two 256-channel headstages.

data rate of up to 90 MBps, with one controller dedicated to the memory card and the other interfacing with the 1LV WiFi module.

The system's data handling capabilities are further enhanced by the Cross-link NX FPGA LIFCL-40 (Lattice Semiconductor), featuring 40K Lookup Tables (LUTs) in a 121-ball, 0.5 mm pitch package. This FPGA is responsible for converting the analog data from the headstages, packing it into blocks of a convenient size, and transmitting these blocks to the MCU via the Octal SPI bus. In certain configurations, the FPGA also performs lossless data compression using differential coding, achieving up to a 2.5x compression ratio. This compression allows for efficient storage of neural data, enabling 1,024 channels of 20 kHz, 10-bit data to be stored on a 1TB memory card for up to 1.15 days.

To accommodate the necessary communication between analog front-ends (3.3V interface) and the MCU (1.8V interface), logical level shifters are employed. All other digital communication on the PCB uses 1.8V, optimizing power efficiency. The 10-layer PCB design, with a thickness of 1 mm, was crucial in achieving a compact and lightweight

form factor. The final board size is 28 mm by 17 mm, and the total weight, including the memory card, is just 3.25 g.

The neural logger has been tested with up to 1,024 channels at 20 kHz, with future tests planned to scale the system to 2,048 and 3,072 channels. In terms of power consumption, the logger draws 123 mA, 192 mA, and 320 mA from a 4V power supply when configured with 256, 512, and 1,024 channels using Intan chips, respectively. For the 1,024-channel configuration with the MEA1k headstage, power consumption is estimated at 180 mA without compression and 124 mA with data compression enabled.

Chapter 3

Results

3.1. Fabrication of UFTEs with flexible connectors

The Ultra-Flexible Tentacle Electrodes (UFTEs) were fabricated at the ETH Zurich cleanroom facility. Each device features four shanks, containing a total of 1,160 contact pads arranged in a two-pad-per-fiber configuration. The design allows for mechanical independence of each fiber in the brain, minimizing tissue damage during long-term implantation. The electrodes are constructed from two layers of Ti/Au (Titanium/Gold) metal, which are sandwiched between three layers of polyimide, a material known for its high bio-compatibility and long-term stability. Each fiber has dimensions of 7 um in width and 3.6 um in thickness, resulting in a total cross-sectional area of 25.2 um^2. The recording pads themselves have a surface area of 13 x 13 um^2.

For attachment to the MEA1k chip, the flexible contact pads have a diameter of 50 um, with spacing dynamically adjusted based on routing requirements. A minimum distance of 22 um is maintained between the contact pads to prevent electrical short circuits. The total number of flexible connection pads between the UFTEs and the MEA1k is 1,278, covering an area of 2.1 mm x 3.85 mm^2. This results in a high-density electrode configuration, with a channel density of 158 channels/mm^2.

The following sections highlight the key stages of the fabrication process, with accompanying optical microscopy images to illustrate each step:

The first metal layer, consisting of 10 nm Titanium and 150 nm gold, was deposited via electron beam evaporation onto a 1 um thick polyimide layer. The metal layer was then encapsulated with another polyimide layer for electrical isolation. Figure 3.1 a shows the structure after the patterning of the Ti/Au layer and subsequent encapsulation. The flexible interface area (Figure 3.1) shows 639 flexible contact pads over a 2.1 mm x 3.85 mm area. Zoomed-in images in Figures 3.1 d,g show individual 50 um diameter

contact pads. The metal traces showed minimal expansion (less than 0.5 um), ensuring a minimum spacing of 2.5 um between gold traces.

The second metal layer follows the same process as the first, consisting of 10 nm Titanium and 150 nm gold. After deposition, the second metal layer is insulated with a third polyimide layer. This step doubles the number of flexible contact pads to 1,278, exceeding the MEA1ks 1,024-channel limit. The dual contact pad arrangement for each fiber, offering enhanced spatial resolution, is shown in Figure 3.2.

Following the fabrication of the second metal layer, Reactive Ion Etching (RIE) was used to remove polyimide from specific areas, such as the contact pads for recording neural activity and the flexible contact pad areas for bonding to the MEA1k surface. The etching also shaped the individual fibers and device boundaries. Figure 3.3 shows the electrode bundles and the precise removal of polyimide from the contact pads after the etching process.

The surface area of the contact pads was then coated with PEDOT:PSS, which reduces the impedance of the electrodes to 283.8 kΩ (74.99 kΩ std).

The electrode contact pads are distributed across four bundles, each customized based on the implantation depth and location. The first shank contains 200 electrode pads100 pairs of two electrode contact padsalong a 5950 um length of the shank. The second shank has 354 electrodes (172 pairs) along a 10,450 um length shank. The third shank includes 356 electrodes (173 pairs) along a 10,450 um length shank. The last shank has 250 channels (125 pairs of electrodes) along a 6200 um length of the shank. The total number of electrode pads sums up to 1160 electrodes (580 pairs). The mechanical independence of each of these shanks enables distributed recording from multiple areas of the brain without the limitation of insertion depth or angle.

The results of the fabrication process in Figures 3.1, 3.2, and 3.3 demonstrate the design and fabrication quality of the electrodes.

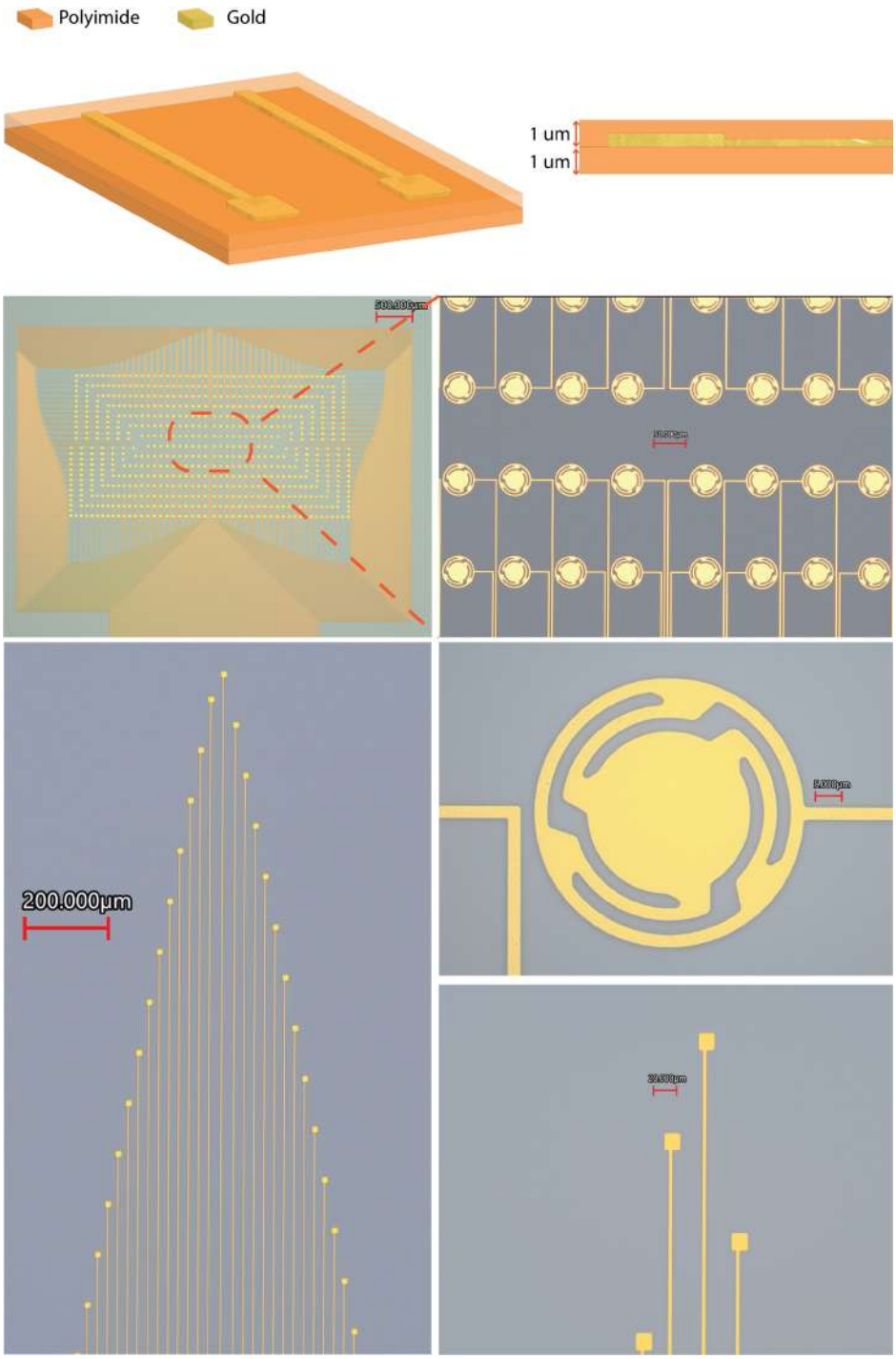


Figure 3.1.: **First Insulated Metal Layer**: (**a**) First insulated metal layer fabrication step (**b**) wide view of the flexible contact area (**c**) Integration of the Flex2chip design with the underlying CMOS-MEA (**d, f**) close-up optical images of the flexible connectors. (**e, g**) contact pads of a single shank.

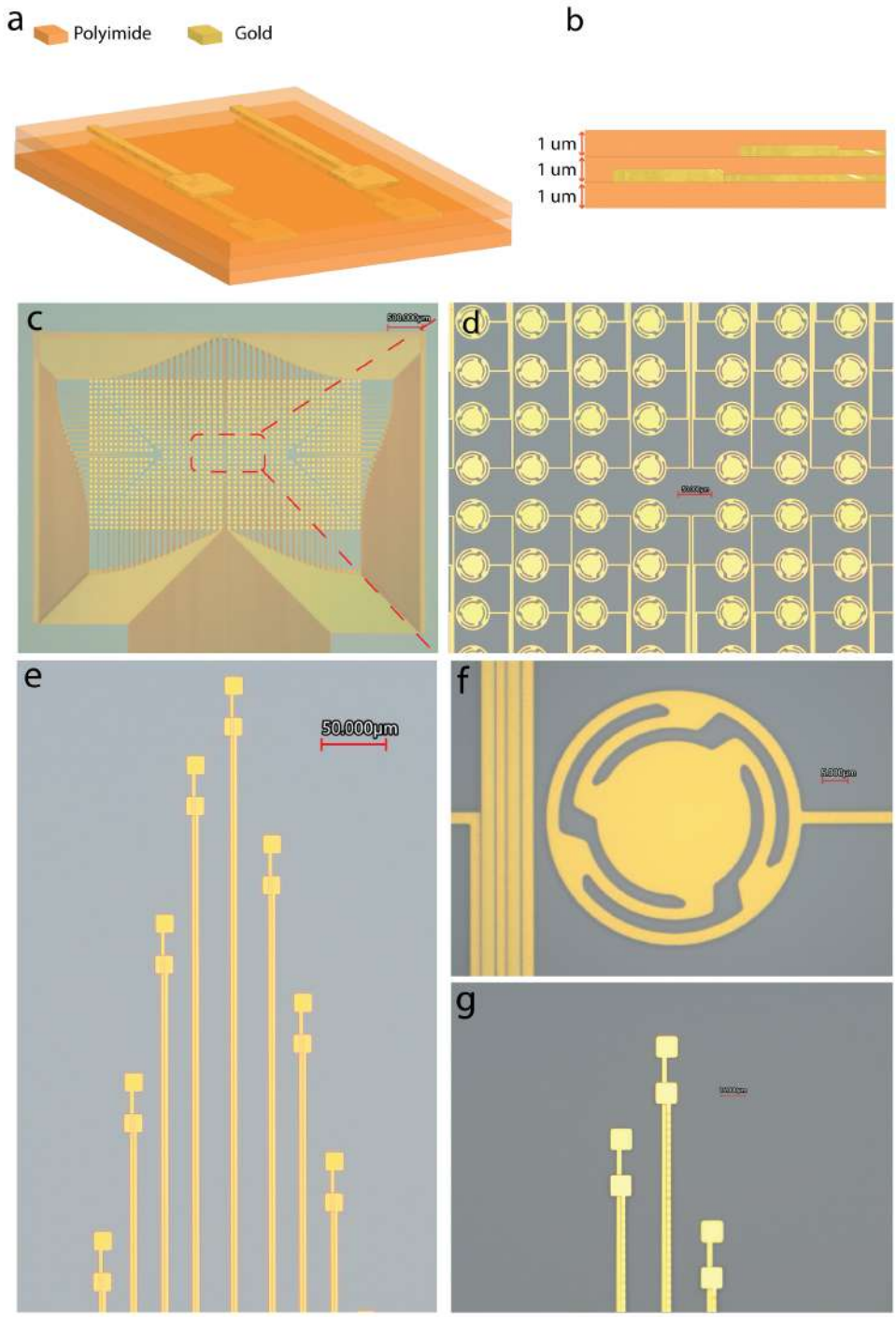


Figure 3.2.: **Second Insulated Metal Layer**: (**a**) Second insulated metal layer fabrication step (**b**) wide view of the flexible contact area (**c**) Flex2chip interface design for bonding to the MEA1k (**d, f**) close-up optical images of the flexible connectors. (**e, g**) dual contact pad arrangement of a single shank.

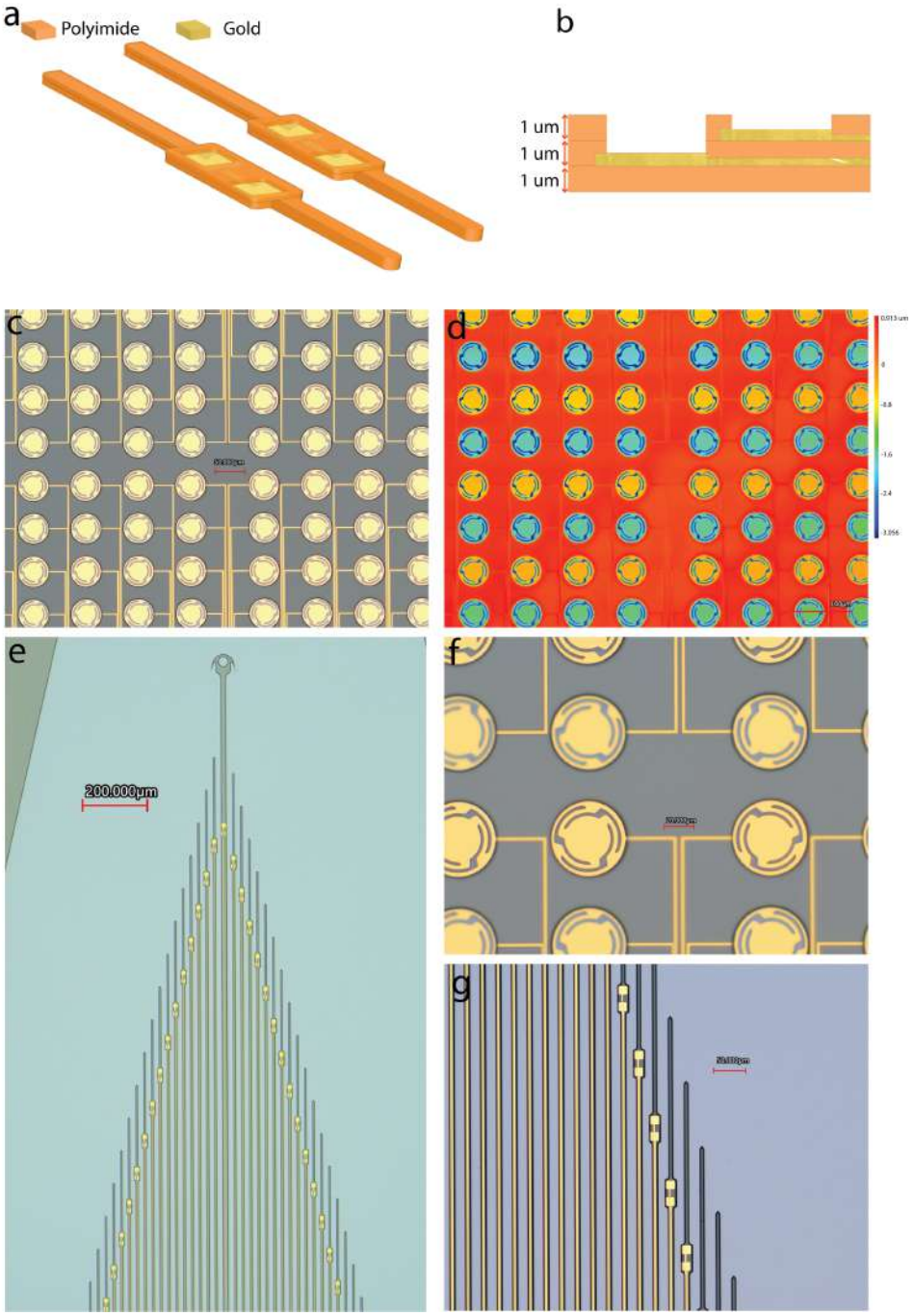


Figure 3.3.: **Etching of the Polyimide Layer**: (**a, b**) Etching step to remove polyimide from the electrode pads, connectors, and shape the fibers (**c**) wide view of the flexible contact area (**d**) surface profile of the connector pads illustrating the multi-layer structure (**f**) close-up optical images of the flexible connectors. (**e, g**) formation of individual fibers along a shank.

3.2. Gold-plating results of the MEA1k

We achieved the desired gold plating thickness of 3.5 um, elevating the MEA1k's recessed pads by 2 um above the surface, which ensured reliable electrical contact with our flexible connectors. The plating was uniform, and there were minimal short circuits between the closely spaced pads (8.2 um apart). By carefully controlling the current, temperature, and other key parameters, we maintained the quality and robustness of the electrodes, essential for long-term neural recording. For more details on the process, please refer to the methods section. As illustrated in 3.5

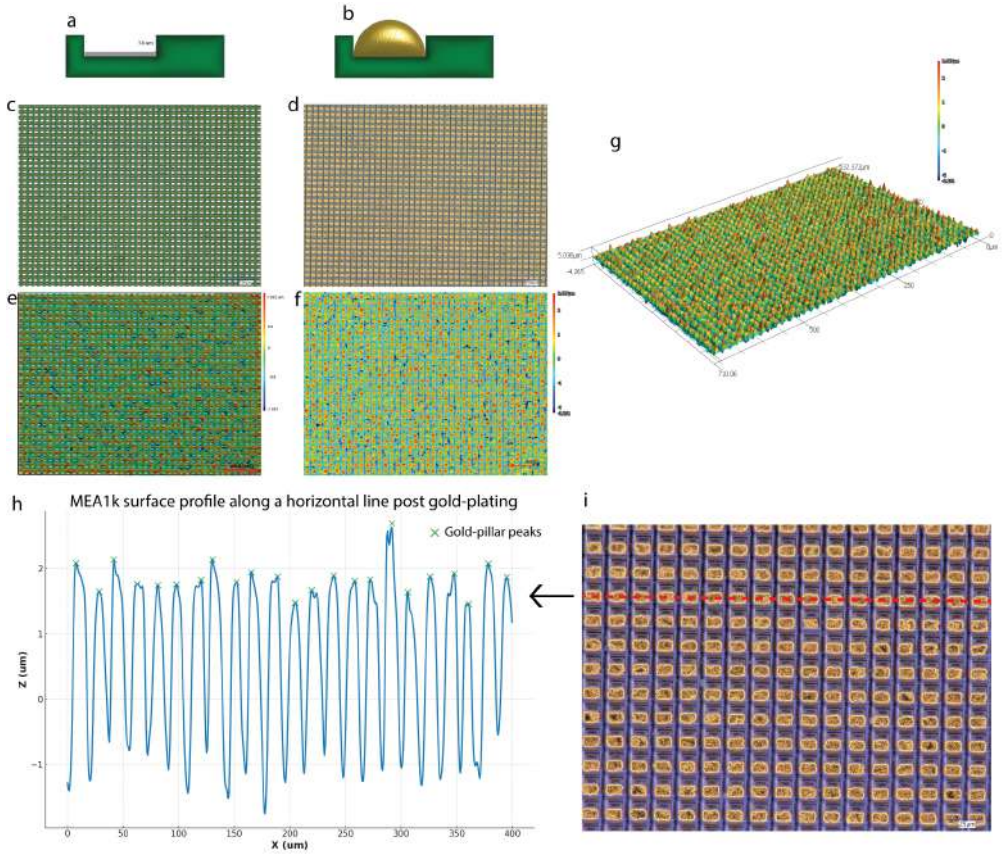


Figure 3.4.: **Gold Plating Results of an MEA1k Chip**: (**a, b**) Schematic illustration of a single MEA1k recording pad before and after the gold plating process. (**c**) A wide view of the MEA1k pads prior to gold plating. (**d**) Wide view of the MEA1k pads after gold plating. (**e**) Surface profile of the pads before the gold plating process. (**f, g**) Surface profile of the pads after gold plating, demonstrating uniform height distribution across the pads. (**h, i**) closae up view of the gold-pillars on top of the MEA1k and the height profile along a horizontal line. Mean peak of the pillars is 1.83 um with 0.18 um std.

3.3. Assembly of the Flexible Electrodes to the MEA1k

After fabrication, the UFTEs were delicately peeled off the silicon wafer by applying water droplets along the edges to facilitate detachment. Care was taken to keep the devices flat during the assembly onto the gold-plated MEA1k to prevent damage to the flexible contact pads. The pads were roughly aligned with the MEA1k borders, and a drop of IPA was applied to assist in alignment. This also helped initiate capillary and Van der Waals forces, providing a stable mechanical connection between the electrodes and the MEA1k surface.

To ensure stability during handling and surgery, a custom titanium holder was designed to secure the electrodes onto the MEA1k pads. A small piece of Ecoflex, a silicone-based material known for its flexibility and biocompatibility, was placed between the holder and the electrode surface to provide soft, uniform contact. Ecoflex, due to its durability and mechanical properties, is ideal for flexible implants and medical devices, offering safe, long-term compatibility with biological tissues.

3.4. Characterization of the UFTE-MEA assembly

The integrity of the connections between the flexible electrodes and the MEA1k pads was evaluated by applying a 5 mV sinusoidal voltage at 1 kHz across all 26,400 pads of the MEA1k array. Channels were selected for further analysis if they recorded at least 80% of the applied signal power. In our bonding test, we achieved a 72.4% yield, successfully connecting 843 flexible pads to the MEA1k recording pads out of the 1,278 available flexible connectors (see Figure 3.7). The results displayed distinct clusters of connected MEA1k pads, each corresponding to a single flexible pad from the electrode side. Notably, having multiple MEA1k pads connected to one flexible electrode pad offered flexibility, allowing us to select the pad with the highest signal connectivity. Importantly, no cross-connections were observed between clusters, confirming the absence of short circuits, which validates the importance of maintaining the minimum 22 um spacing between flexible pads for optimal clustering. The non-connected flexible pads were primarily attributed to misalignment between the flexible device and MEA1k pads, as well as the narrow placement of epoxy on the MEA1k surface, which introduced upward forces and prevented proper contact between some pads and the MEA1k chip.

We further assessed the long-term stability of the connections by monitoring the connectivity of a flexible electrode assembled onto the MEA1k chip over several weeks. The results demonstrated that the flexible pads maintained stable connections to the same MEA1k pads with consistent connectivity values throughout the experiment. Repeated testing revealed a stable distribution pattern with slight improvements in connectivity over time, likely due to the evaporation of isopropyl alcohol (IPA) initially used to establish contact. No electrode connections were lost during the test period, confirming the systems reliability for long-term studies spanning weeks to months.

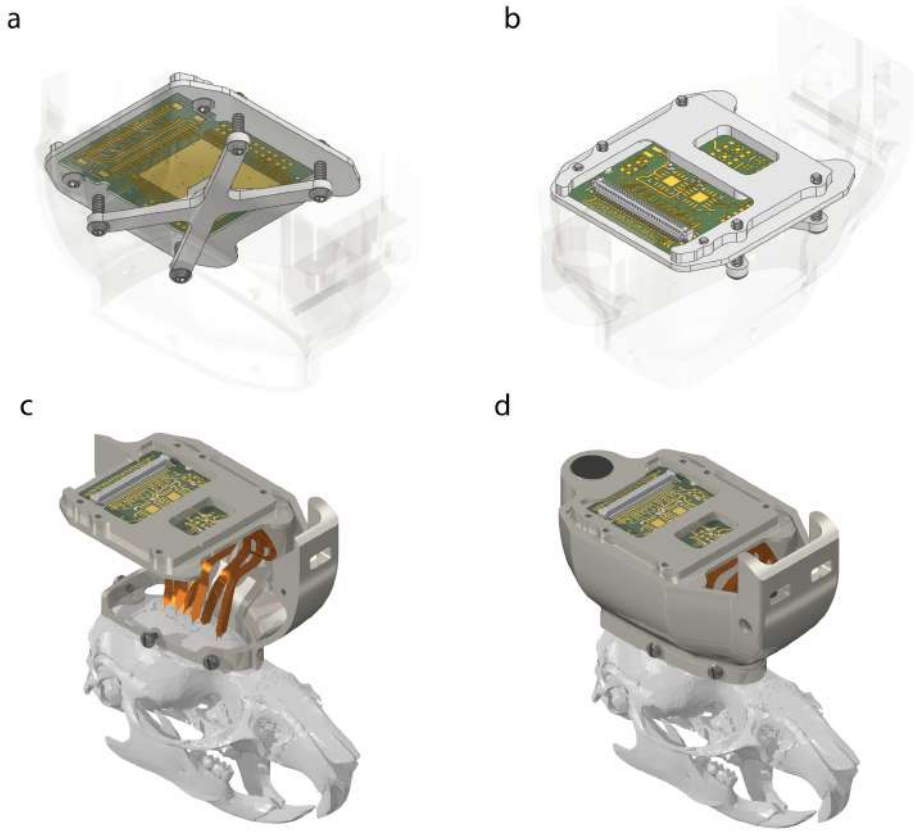


Figure 3.5.: **Assembled MEA1k with Titanium Holder on the Animal's Headstage**: (**a, b**) Top and bottom views of the custom MEA1k PCB secured to the titanium holder. (**c, d**) Diagrams showing the placement of the assembled PCB on the animal's headstage, illustrating the precise positioning of the electrodes as they penetrate the brain.

Finally, impedance measurements of individual electrodes were conducted by applying an external sinusoidal current of 2 uA at 1 kHz to the electrodes. The measured impedance values averaged 283.8 kΩ (74.99 kΩ std), consistent with previous results from our UFTEs paper [1]. Differences in impedance values can be attributed to the different methodologies employed for measurement. In the Intan-based system, the Intan chip itself injects a 1 nA current into each electrode channel and measures the resulting voltage to calculate impedance. However, since the MEA1k does not have this functionality, we employed an external current generator to measure the voltage for each electrode. A potential limitation of this approach is the possibility of current leakage through other channels, which may affect the accuracy of the impedance measurement. To mitigate this issue, we routed all 1,023 unused analog-to-digital converters to channels with the lowest connectivity values, ensuring minimal current leakage during the

measurements.

This approach allowed us to confirm the robustness of the electrode-MEA1k bonding, as well as the reliable impedance characteristics necessary for long-term neural recording experiments. We then moved on with the Surgery to implant the system into a rodent model.

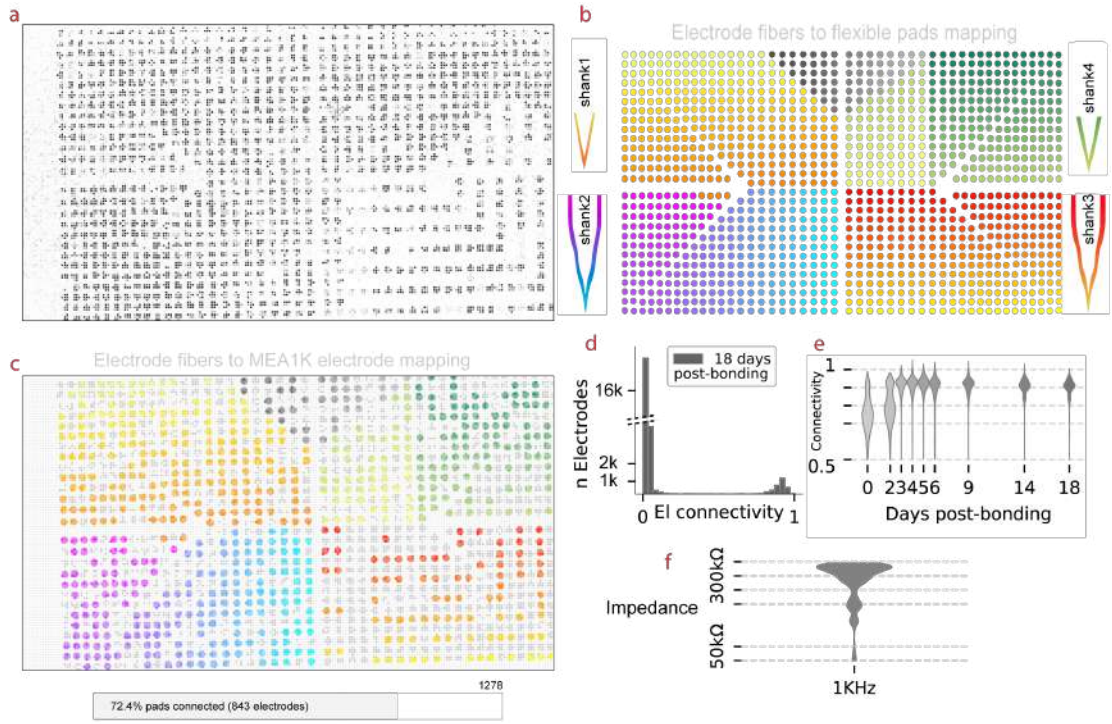


Figure 3.6.: **Connectivity Analysis and electrical characteristics of the UFTE-MEA assembly**: (**a**) Connectivity map of the MEA1k pads post-bonding (**b**) Mapping of the Electrode fibers of each shank to the flexible contact pads interface. (**c**) Alignment of the electrode mapping to the connectivity results of **a**, only the channels with minimum 0.8 connectivity value were kept for analysis. (**d, e**) Long-term stability of the systems up to 18 days post bonding (longest duration tested). **f** Impedance measurement of the electrode fibers with 283.8 kΩ avg. (74.99 kΩ std.)

3.5. In-vivo experiment results

The surgical implantation of the electrodes was conducted on Thursday, October 17th, 2024, with two out of the four shanks successfully implanted into the brain. One shank was damaged during the lift-off process from the silicon wafer, which compromised the hook at the tip of the electrode bundle, while the other failed due to the curvature introduced after the silk fibroin coating, making insertion problematic. However, shank 1, with 200 channels, and shank 2, with 356 channels, were properly implanted. Based

on prior surgical experiences, it is expected that animals require 2 to 6 days to recover, with one week post-surgery being the optimal time for neural recordings to capture single-unit activity with a high signal-to-noise ratio, as brain tissue requires time to heal post-implantation.

To provide preliminary in vivo results, a recording session was performed four days post-implantation. Pre-selected MEA1k pads, based on the connectivity results of the MEA-UFTE bonding, were routed for recording. Minimal noise was detected across the selected channels, and no evidence of the 50 Hz noise observed in prior in vivo experiments was present. Even at this early stage, several channels exhibited clear single-unit activity in the raw traces, as seen in the example below.

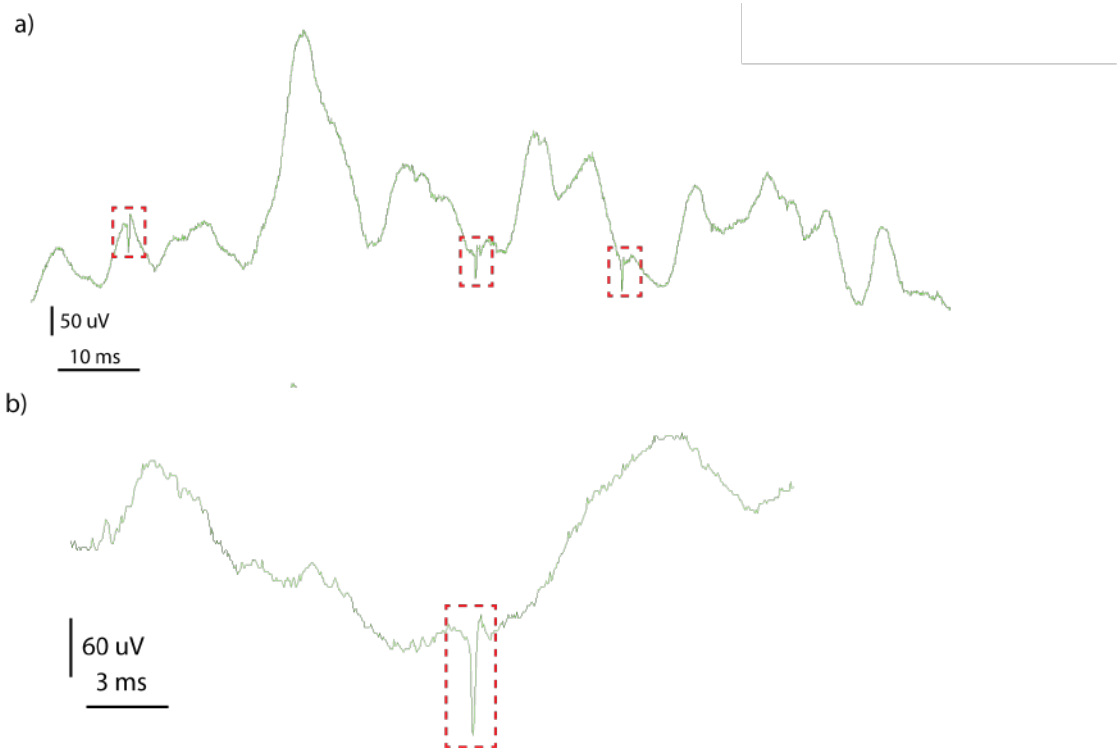


Figure 3.7.: **Raw traces of two MEA1k channels, four days post-surgery recording session**: (**a**) Multiple spikes observed in the raw trace of an electrode from the hippocampus. (**b**) Zoomed-in view of a single-unit activity recorded in one of the channels.

Despite the short recovery period before this recording session, single-unit activity was clearly visible, underscoring the system's capability to detect neural signals with minimal background noise. Future recordings are anticipated to yield even clearer signals with improved signal-to-noise ratios as the brain tissue fully heals, which typically occurs by one week post-surgery.

Chapter 4

Conclusion

We have developed a high-density neural recording system that integrates the long-term stability and durability of Ultra-Flexible Tentacle Electrodes (UFTEs) with the high-density recording capabilities of CMOS-based microelectrode arrays (MEAs). By utilizing Flex2chip structures to interconnect UFTEs with a MEA1k chip, we achieved an interconnection density of 158 channels/mm^2. This advancement enabled the design of a new neural recording system that reduces the printed circuit board (PCB) space by 40% while increasing the number of recorded channels by a factor of four, from 256 to 1,024 channels, without the need for stacking multiple chips on the animals head-stage. The UFTE-MEA system demonstrated stable, long-term connections between flexible electrodes and the MEA1k recording pads, maintaining low impedance values. This stability was further confirmed by in vivo recordings, which showed clear single-unit activity, validating the system's recording capabilities.

To achieve this multi-thousand-electrode density within an 8 mm^2 surface area, several key enhancements to the Flex2chip design were implemented. First, a multi-layer metallization design reduced routing space by 50%, allowing the integration of 1,278 flexible pads with a 50 um diameter within the MEA1k surface area, compared to the 720 channels achieved with 40 um diameter pads in previous designs [2]. Despite the increased number of channels, a 23 um minimum spacing between flexible pads was maintained to prevent short circuits between neighboring electrodes. This design resulted in well-isolated clusters of MEA1k pads connected to a single flexible electrode, with no crosstalk between electrodes. The system also demonstrated its ability to support intracortical recordings with multiple shanks, independent of insertion depth or angle, making it highly versatile for chronic, in vivo recordings.

Further refinements included the development of a custom PCB for the MEA1k chip to ensure compatibility with the custom-designed animal head-stage, as well as gold-plating the MEA1k recording pads to elevate their surface by 1.8 um, ensuring robust physical

contact with the UFTEs. The electrodes were securely positioned using a custom titanium holder, and connectivity testing revealed no degradation in the electrode-MEA1k bonds over several weeks, confirming the long-term stability of the system.

Following these assessments, the system was surgically implanted into an animal model. Initial recordings were conducted three days post-surgery, detecting single-unit activity across multiple channels, even though optimal recordings are typically expected one week post-surgery after tissue recovery. These preliminary results indicate the system's effectiveness, with further assessments planned to evaluate noise levels and single-unit yields once the tissue has fully healed.

In addition to the tethered recording system, a wireless neural recording system was introduced, leveraging the MEA-UFTE integration. A custom-designed neural logger replaced the host computer, allowing data acquisition and storage from 1,024 channels at a rate of 25.6 MBps, with a compact form factor weighing only 13.84 grams, making it suitable for small laboratory animals. The logger can support up to three MEA1k boards in parallel, enabling simultaneous recording from up to 3,072 channels in freely moving animals.

Overall, this advancement in the integration of UFTEs with high-density MEAs significantly enhances the scalability of electrophysiological recordings, facilitating large-scale neural data acquisition from multiple brain regions in both tethered and wireless configurations.

Appendix A

Declaration of Originality

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Authored by:
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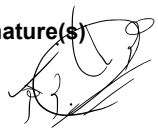
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[2] E.g. ChatGPT, DALL E 2, Google Bard
[3] E.g. ChatGPT, DALL E 2, Google Bard

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